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Fast and robust DC link voltage control of single-phase grid-connected converters supporting 50 Hz and 60 Hz mains with universal magnitudes range☆

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ABSTRACT

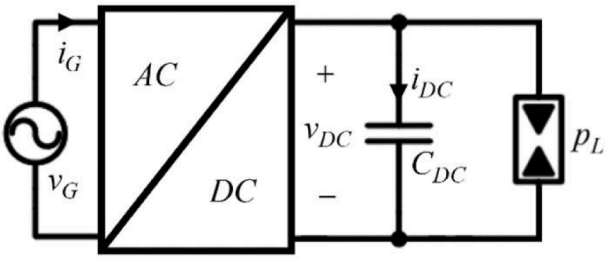
It is well-known that dynamics of single-phase grid-connected power converters employing PI or type-II DC link voltage controllers is limited by the maximum tolerable value of utility-side current total harmonic distortion (THD). Moreover, universal DC link voltage controller must be capable of supporting both 50 Hz and 60 Hz utilities and universal range of magnitudes ($85V_{rms}$–$265V_{rms}$). As a result, DC link crossover frequency value (and hence corresponding settling time) may vary significantly in case grid voltage magnitude variations are not compensated in the controller. In order to cope with the above-mentioned deficiencies, the paper suggests a design methodology for a novel dual-frequency-tuned utility magnitude insensitive plug-in disturbance observer (DOB), supporting universal utility frequencies and magnitudes so that the transient response of a grid-connected converter is improved without altering the existing PI or type-II DC link voltage controller coefficients. These characteristics are attained by robust modification of DC link voltage loop plant to possess notch-like properties around two frequencies of interest and treating the grid voltage magnitude value as non-measurable uncertainty translated into input disturbance and subsequently cancelled. Analytical design guidelines for obtaining PI + DOB controller coefficients are provided for desired values of utility-side current THD and DC link voltage loop phase margin (PM). Simulations and experimental results are provided to validate the feasibility of proposed design methodology.

1. Introduction

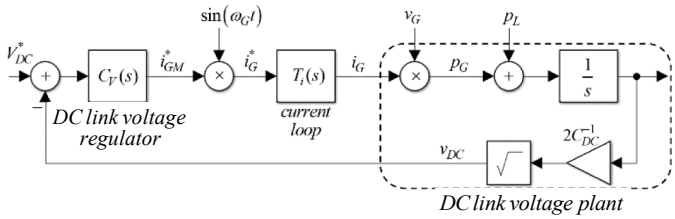
Modern power quality standards imply sinusoidal-shaped current exchange [1] between grid-connected converter and the utility [2]. Such operation leads to pulsating instantaneous power flow, formed by DC and double-grid-frequency components [3]. As a result, capacitive DC links are employed to absorb the pulsating power component [4] in addition to compensating the mismatch between grid-side and load-side instantaneous power [5] during transients. At the same time, average value of the voltage across DC link capacitor terminals is regulated to a constant reference [6] (typically by either PI or type-II compensator [7]) in order to ensure steady-state system energy balance [8]. Consequently, pulsating current flowing through the DC link capacitor gives rise to double-grid-frequency DC link voltage ripple proportional to apparent power processed by the converter and inversely proportional to DC link capacitance value [9]. When sensed and fed back to DC link regulation loop, DC link voltage ripple component distorts the grid-current reference, increasing corresponding THD value [10]. Therefore, magnitude of DC link voltage loop gain around double-grid-frequency should be limited [11]. In case PI or type-II compensator is employed as DC link voltage regulator [12], corresponding loop gain magnitude is monotonically decreasing and trade-off between AC-side current THD and DC link voltage dynamics appears [13]. In order to decouple the above two performance merits, it was suggested [14] to introduce a double-grid-frequency tuned notch filter into the DC link loop [15] while re-designing the existing PI or type-II compensator [16]. An alternative approach for designing the notch filter was suggested in [17] in order to enhance transient responses at both the DC bus voltage and the output current. Notch-filter-based approach was recently extended to support harmonic current exchange between the converter and the utility by introducing several notch filters tuned to relevant multiples of double-grid-frequency [18] and elaborated in [19] by adding adaptation

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(a) Generalized topology.



(b) DC link voltage loop structure.

Fig. 1. Single-phase grid-connected power converter.

mechanism to allow precise tracking of the main frequency. Still, redesign of PI or type-II compensator would be necessary to increase the attainable DC link crossover frequency. In order to eliminate the necessity for existing DC link voltage regulator redesign while enhancing the dynamic response of the converter without violating the maximum allowed utility-side current, plug-in DOB-based solution [20] was recently proposed in [21], where an additional degree-of-freedom was used to enhance sensitivity function gain around double-grid-frequency while increasing the overall DC link voltage loop gain magnitude without modifying the existing feedback controller [22]. Unfortunately, neither notch-filter-based solutions [14–19] nor the methodology in [21] took into account the fact that universal grid-connected converter must be capable of supporting both 50 Hz and 60 Hz grid frequencies as well as wide range of possible utility magnitudes, focusing on single-frequency cases and a priori known magnitude value. Consequently, this paper aims to bridge this gap by proposing a design of plug-in DOB possessing similar structure to the one proposed in [21] yet derived from a totally different line of thinking. While the DOB in [21] reshaped the DC link voltage controller by imposing higher controller gain at all but double-grid frequencies, the proposed algorithm reshapes the plant, "seen" by the existing PI or type II controller by imposing constant gain and dual-notch-like behavior at frequencies of interest. Since the grid magnitude directly influences the DC link voltage plant gain, it is treated as uncertainty which is estimated and subsequently cancelled by the DOB. As a result, a system operating under the proposed control algorithm attains a much higher crossover frequency than a classical one and is nearly-insensitive to grid voltage magnitude value. Guidelines for classical PI and proposed PI + DOB controllers design accompanied by analytical expressions of coefficients are provided, based on the desired values of DC link voltage loop phase margin and grid-side current total harmonic distortion. Extensive simulation results and particular operating point experiments are provided to accurately verify the feasibility of the proposed methodology.

The rest of the paper is organized as follows. Essentials of single-phase grid-connected converters design and operation are brought forward is Section II. The proposed DOB-based DC link voltage bandwidth increase and uncertainty reduction are revealed in Section III. Simulations and experiments of the proposed methodology applied to power factor correction front end are discussed in Section IV. The paper is concluded in Section V.

2. Single-phase grid-connected converter essentials

A general single-phase grid-connected converter shown in Fig. 1(a) is typically coupled to a downstream converter (represented by power element p_L [23]) via DC link capacitance C_{DC}. For the sake of brevity (yet without losing generality), the grid connected is assumed to operate with unity power factor while the downstream converter is considered to interface a DC source or load. Moreover, both converters are assumed ideal in terms of efficiency so that the subsequent discussion is valid for both from-grid and to-grid energy flow directions.

In case of near-unity power factor operation, utility-side voltage and current may be approximated by

$$v_G(t) \approx v_{GM}(t)\sin\left(\int \omega_G(t)dt\right) \tag{1}$$

and

$$i_G(t) \approx i_{GM}(t)\sin\left(\int \omega_G(t)dt\right) \tag{2}$$

respectively, with $\omega_G(t)$ [rad/s], $v_{GM}(t)$ and $i_{GM}(t)$ signifying instantaneous mains frequency, voltage and current magnitudes, respectively. The value of mains voltage magnitude resides within a range given by

$$V^{\min}_{GM} \leqslant v_{GM}(t) \leqslant V^{\max}_{GM} \tag{3}$$

while the mains frequency may attain values given by

$$\omega_G(t) = \alpha(t)\cdot\begin{cases}\omega^{50}_{Gn}, & 50Hz\ utility\\ \omega^{60}_{Gn}, & 60Hz\ utility\end{cases} \tag{4}$$

with $\omega^{50}_{Gn} = 100\pi$ rad/s and $\omega^{60}_{Gn} = 120\pi$ rad/s denoting nominal mains frequency values 50 Hz and 60 Hz utilities, respectively, and $\alpha(t) \in [\alpha_{\min} < 1,\ \alpha_{\max} > 1]$ representing corresponding deviation from nominal values. Utility-side instantaneous power is then given by

$$p_G(t) = v_G(t)i_G(t) = \underbrace{0.5v_{GM}(t)i_{GM}(t)}_{p_{GM}(t)} \underbrace{-0.5v_{GM}(t)i_{GM}(t)\cos\left(2\int \omega_G(t)dt\right)}_{\Delta p_{GM}(t)}. \tag{5}$$

Consequently, instantaneous power balance of the system in Fig. 1(a) is defined by

$$v_{DC}(t)i_{DC}(t) \triangleq p_{DC}(t) = p_G(t) - p_L(t). \tag{6}$$

DC link voltage dynamics is then given by

$$v_{DC}(t)\underbrace{C_{DC}\dot{v}_{DC}(t)}_{i_{DC}(t)} = \frac{d}{dt}\left(\frac{1}{2}C_{DC}v^2_{DC}(t)\right) = v_G(t)i_G(t) - p_L(t) \Rightarrow v_{DC}(t)$$
$$= \sqrt{\frac{2}{C_{DC}}\int (v_G(t)i_G(t) - p_L(t))dt} \tag{7}$$

represented by "DC link voltage plant" in Fig. 1(b). Grid-connected converters usually utilize a cascaded controller structure with outer (voltage) loop compensator (denoted as $C_V(s)$ in Fig. 1(b)) goal to preserve the system energy balance by calculating the utility-side current magnitude reference $i^*_{GM}(t)$ required to regulate the average value of instantaneous DC link voltage $v_{DC}(t)$ to a constant set point V^*_{DC}. Unity power factor operation is attained by creating grid-side current reference $i^*_G(t)$ from multiplying $i^*_{GM}(t)$ with utility-voltage-template sin $(\int \omega_G(t)dt)$ and tracking it by means of the inner (current) loop. Since the current loop bandwidth (limited by sampling and switching frequency) is typically much higher than that of the voltage loop (limited by the lowest double-multiple of utility frequency), complementary sensitivity function of the current loop $T_i(s)$ may be accurately approximated as unity within the voltage loop bandwidth so that

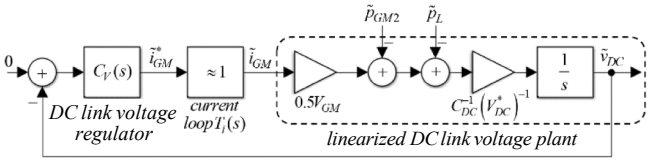


Fig. 2. Linearized DC link voltage loop structure.

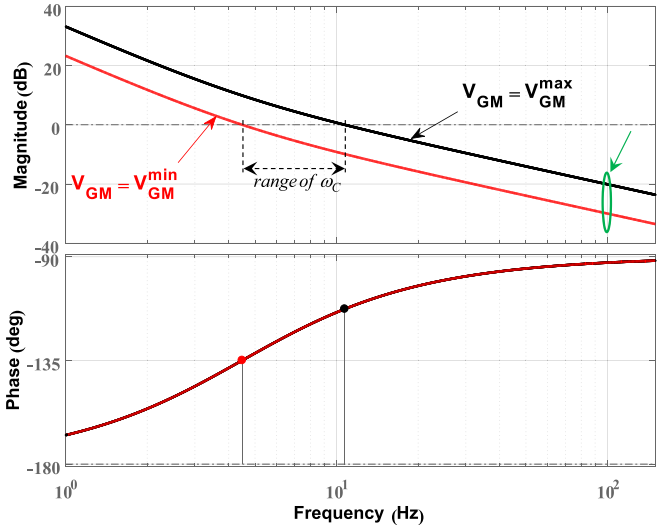


Fig. 3. Bode diagram of DC link voltage loop gains under PI control for $PM^* \geq \pi/4$ rad, $THD^* \leq 0.05$.

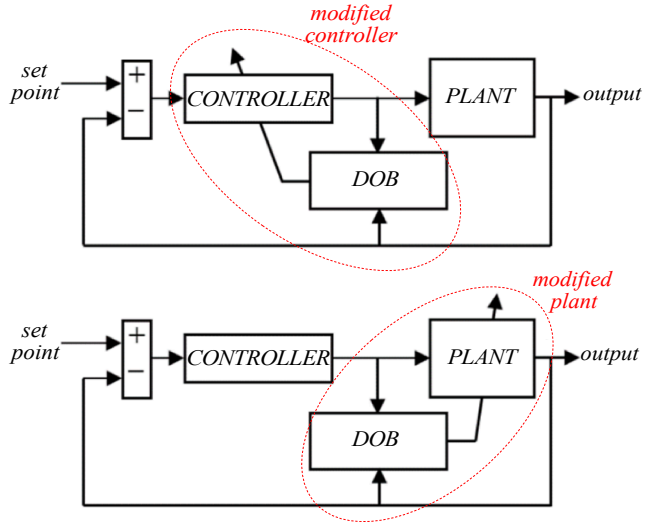


Fig. 4. Two ways of DOB-based additional degree-of-freedom introduction.

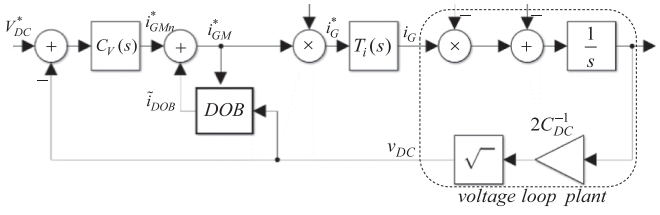


Fig. 5. DOB-assisted DC link voltage loop structure.

$$\frac{i_G(s)}{i_G^*(s)} \triangleq T_i(s) \approx 1 \Rightarrow i_G(t) \approx i_G^*(t) \tag{8}$$

is assumed afterwards in Fig. 1(b).

In steady state $v_{GMn}(t) = V_{GMn}$, $i_{GM}(t) = I_{GM}$ and $p_L(t) = P_L$ are constant with (cf. (5), (6)

$$P_{GM} = P_L \Rightarrow i_{GM}(t) = I_{GM} = \frac{2P_L}{V_{GM1}}. \tag{9}$$

In case the average value of $v_{DC}(t)$ is regulated V_{DC}^*, (7) becomes [24]

$$v_{DC}^{ss}(t) \approx V_{DC}^* - \underbrace{\Delta V_{DC} \sin(2\omega_G t)}_{\Delta v_{DC}(t)} \tag{10}$$

with

$$\Delta V_{DC} \triangleq \frac{P_L}{2\omega_G V_{DC}^* C_{DC}} << V_{DC}^*. \tag{11}$$

Using (10), (11) with Fig. 1(b) yields [25]

$$i_G^{ss}(t) \approx \frac{2P_L}{V_{GM}} \left(\sin(\omega_G t) - \frac{V_{GM}|C_V(2\omega_G)|}{8\omega_G V_{DC}^* C_{DC}} \sin(\omega_G t - \arg C_V(2\omega_G)) + \frac{V_{GM}|C_V(2\omega_G)|}{8\omega_G V_{DC}^* C_{DC}} \sin(3\omega_G t + \arg C_V(2\omega_G)) \right), \tag{12}$$

i.e. the first and the third harmonics dominate the utility-side current (without considering switching-frequency components). Corresponding utility-side current THD may be approximated as [11]

$$THD \triangleq \frac{\left|i_G^{ss}(\omega) - i_G^{ss}(\omega_G)\right|}{\left|i_G^{ss}(\omega_G)\right|} = \frac{\left|i_G^{ss}(3\omega_G)\right|}{\left|i_G^{ss}(\omega_G)\right|} \cong \frac{V_{GM}}{8\omega_G V_{DC}^* C_{DC}} |C_V(2\omega_G)|. \tag{13}$$

Consequently, given the desired total harmonic distortion value THD^* the controller gain must satisfy [11]

$$|C_V(2\omega_G)| \leqslant \frac{8V_{DC}^* C_{DC} THD^* \omega_G}{V_{GM}}. \tag{14}$$

Combining with (3), (4), the constraint (14) may be narrowed as

$$|C_V(2\omega_G)| \leqslant 8V_{DC}^* C_{DC} THD^* \frac{\omega_G}{V_{GM}^{\max}}, \quad \begin{matrix} \alpha_{\min}\omega_{Gn}^{50} \leqslant \omega_G \leqslant \alpha_{\max}\omega_{Gn}^{50} \\ \alpha_{\min}\omega_{Gn}^{60} \leqslant \omega_G \leqslant \alpha_{\max}\omega_{Gn}^{60} \end{matrix} \tag{15}$$

due to the fact that for a given power load the highest utility voltage magnitude implies the lowest utility current magnitude. Furthermore, it is interesting to note that if a DC link regulator with monotonically decreasing magnitude (e.g. PI or type-II) is employed, then (15) may be reduced to

$$\left|C_V(2\alpha_{\min}\omega_{Gn}^{50})\right| \leqslant \frac{8V_{DC}^* C_{DC} THD^* \alpha_{\min}\omega_{Gn}^{50}}{V_{GM}^{\max}} \tag{16}$$

since such a controller possesses the maximum gain at the lowest frequency of interest.

Linearizing (7) yields small-signal DC link voltage dynamics (neglecting the dynamics of slow-varying utility voltage magnitude and frequency) given by

$$\frac{d\tilde{v}_{DC}(t)}{dt} = \frac{1}{C_{DC}V_{DC}^*} \left(0.5V_{GM}\tilde{i}_{GM}(t) - \left(\tilde{p}_L(t) + \tilde{p}_{GM2}(t) \right) \right) \tag{17}$$

with

$$\tilde{p}_{GM2}(t) \triangleq \tilde{p}_{GM}(t)\cos(2\omega_G t), \tag{18}$$

represented by "linearized DC voltage loop plant" in Fig. 2.

Defining corresponding plant transfer function as

$$P_V(s) \triangleq \frac{\tilde{v}_{DC}(s)}{\tilde{i}_{GM}(s)} = \underbrace{\frac{V_{GM}}{2C_{DC}V_{DC}^*}}_{G_V} \frac{1}{s} = \frac{G_V}{s} \tag{19}$$

there is

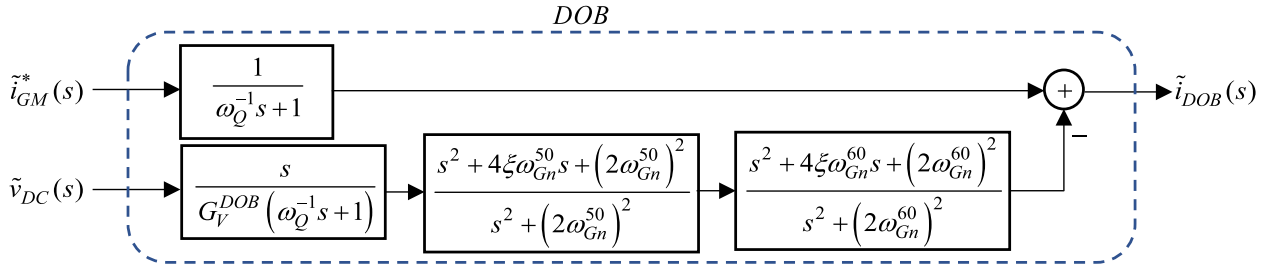


Fig. 6. Graphical illustration of DOB in Fig. 5 implemented according to (46).

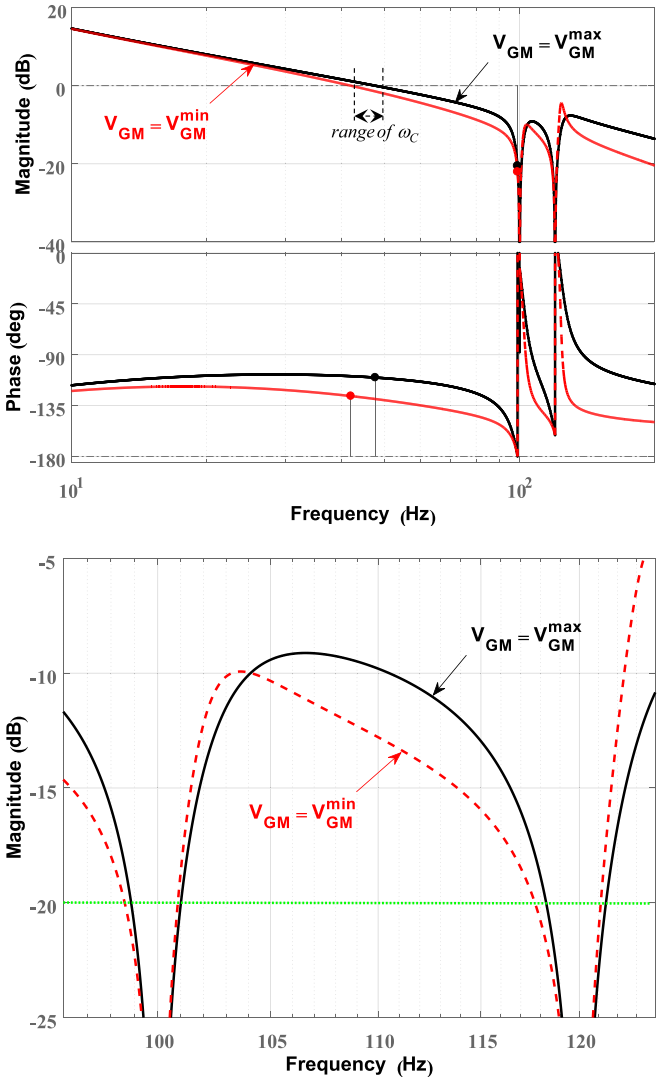


Fig. 7. Bode diagram of DC link voltage loop gains under PI + DOB control for $PM^* \geq \pi/4$ rad, $THD^* \leq 0.05$.

$$|P_V(2\omega_G)| = \frac{G_V}{2\omega_G} \tag{20}$$

and (14) may be rewritten as

$$|C_V(2\omega_G)| \leqslant \frac{2\cdot THD^*}{|P_V(2\omega_G)|}. \tag{21}$$

Consequently, (21) may be satisfied for given THD^* by decreasing either the controller gain or the plant gain within the frequency band covered by $2\omega_G$ (cf. (4)). While controller gain reduction is the classical way of complying with (21), the proposed DOB-based control employs the second option by manipulating the plant "seen" by the controller, as shown next. Furthermore, DC link voltage loop gain given by

$$L_V(s) \triangleq C_V(s)P_V(s) \tag{22}$$

must satisfy (cf. (21))

$$|L_V(2\omega_G)| \leqslant 2\cdot THD^*, \tag{23}$$

e.g.

$$|L_V(2\omega_G)| \leqslant 0.1 \tag{24}$$

for practical values of $THD^* < 0.05$. If a DC link regulator with monotonically decreasing magnitude (e.g. PI or type-II) is employed, then (24) implies loop gain crossover frequency

$$\omega_C \triangleq \omega_{|L_V(\omega)|=1} \tag{25}$$

to reside way below $2\omega_G$, imposing relatively poor dynamics by means of slow transient response to load power variations. In order to accurately quantify the attainable crossover frequency, consider a classical PI controller

$$C_V(s) = K\frac{\tau s+1}{s} \tag{26}$$

as DC link voltage loop regulator. Combining with (16) and (19) yields the following constraint,

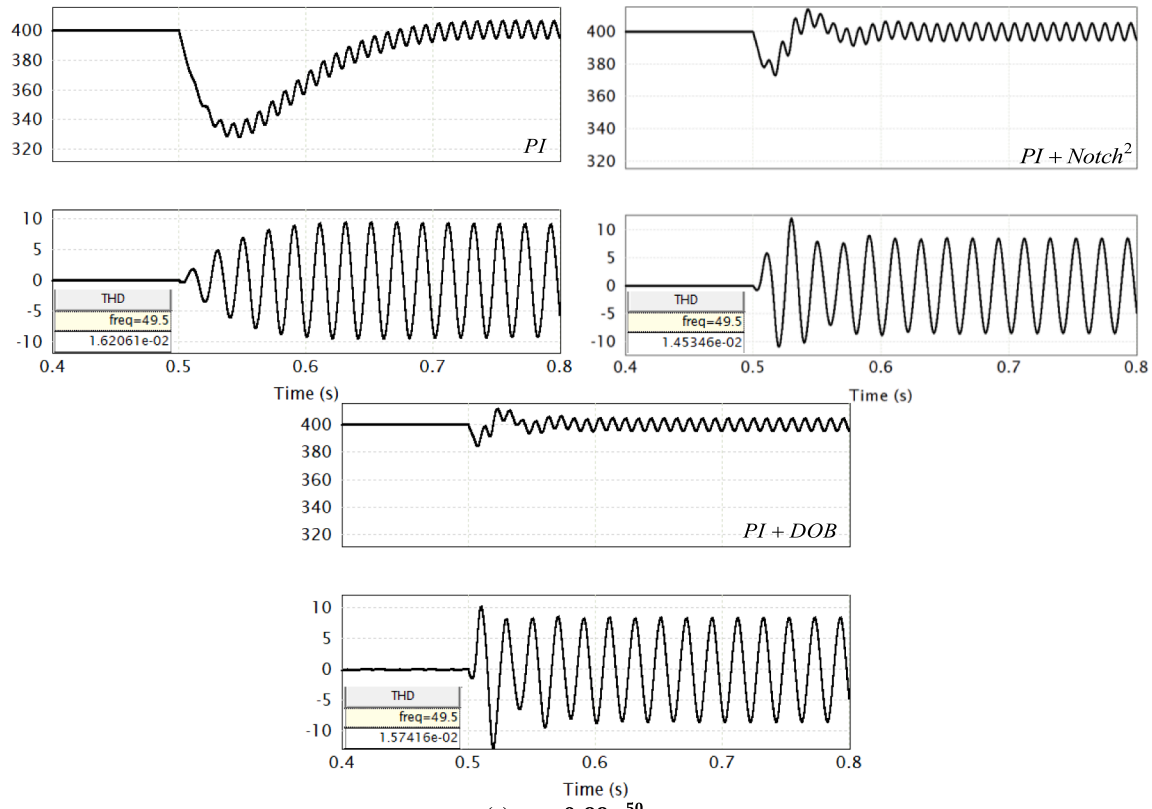
$$KG_V|_{V_{GM}=V_{GM}^{max}} \frac{\sqrt{\tau^2\left(2\alpha_{\min}\omega_{Gn}^{50}\right)^2+1}}{\left(\alpha_{\min}\omega_{Gn}^{50}\right)^2} \leqslant 2\cdot THD^*. \tag{27}$$

Corresponding loop gain is given by

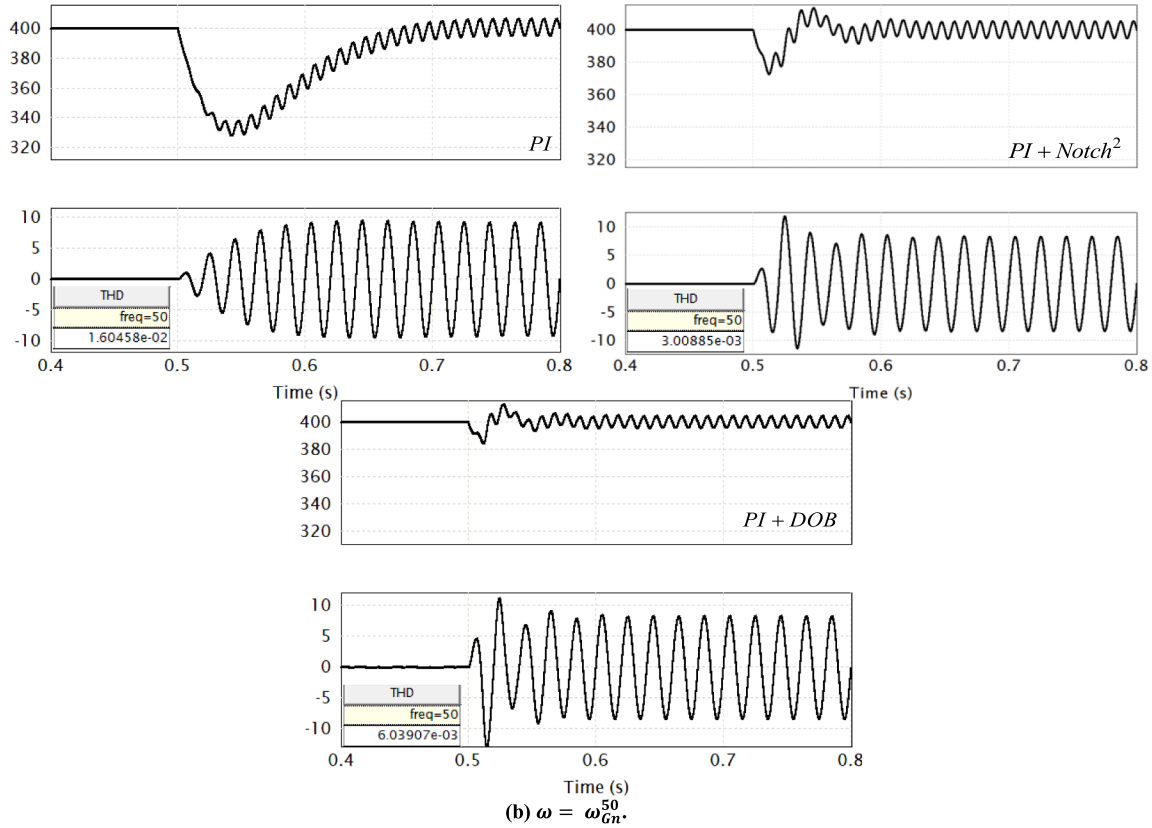
$$L_V(s) = KG_V\frac{\tau s+1}{s^2}. \tag{28}$$

At the crossover frequency (25) the following holds,

$$|L_V(\omega_C)| = KG_V\frac{\sqrt{\tau^2\omega_C^2+1}}{\omega_C^2} = 1 \Rightarrow \omega_C^4 - (KG_V)^2\tau^2\omega_C^2 - (KG_V)^2 = 0 \Rightarrow \omega_C = \frac{KG_V\tau}{\sqrt{2}}\sqrt{1+\sqrt{1+\frac{4}{(KG_V)^2\tau^4}}}$$
$$\arg L_V(\omega_C) = tg^{-1}(\omega_C\tau) - \pi = -\pi + PM \Rightarrow \omega_C = \frac{tg(PM)}{\tau}. \tag{29}$$

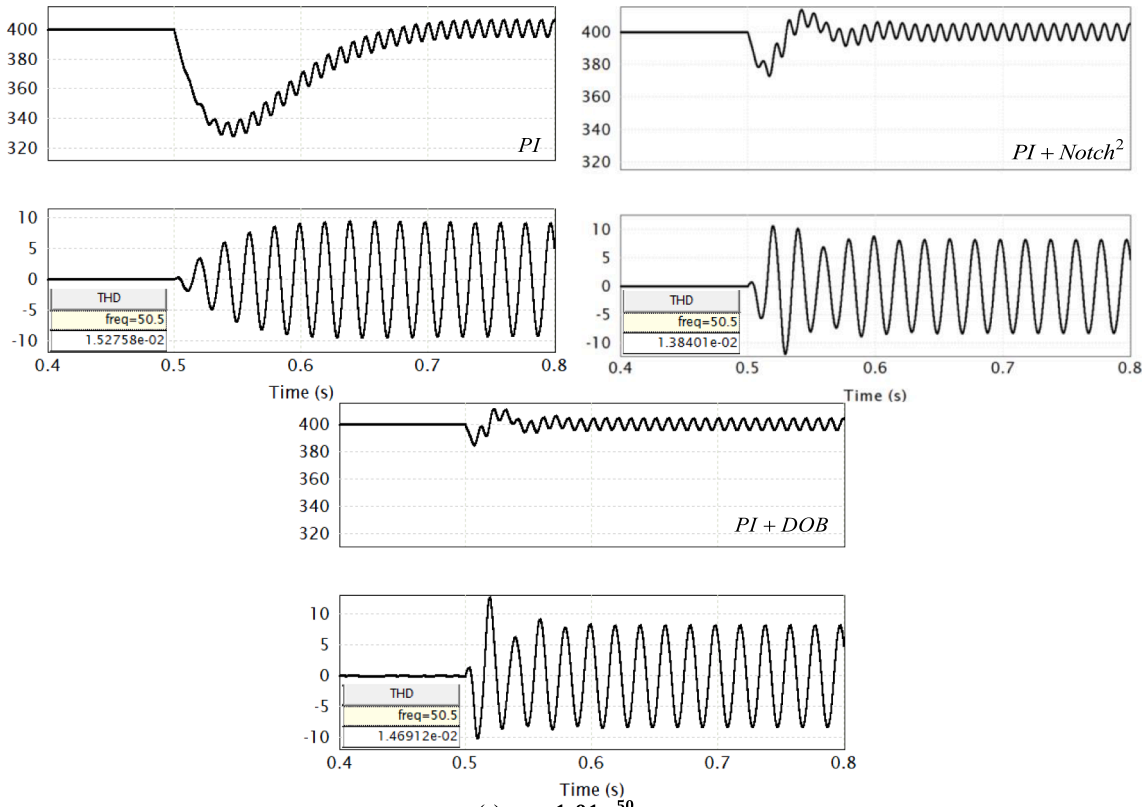


(a) $\omega = 0.99\omega_{Gn}^{50}$.



(b) $\omega = \omega_{Gn}^{50}$.

Fig. 8. Simulation results: DC link voltage (top) and grid-side current (bottom) responses to 0-to-500 W load step around ω_{Gn}^{50} for $V_{GM} = 85\sqrt{2}V$.



(c) $\omega = \mathbf{1.01}\omega_{Gn}^{50}$.

Fig. 8. (*continued*).

Solving (29) for the minimum value of G_V given the desired phase margin value PM^* yields

$$KG_V|_{V_{GM}=V_{GM}^{\min}}\frac{\sqrt{tg^2(PM^*)+1}}{tg^2(PM^*)}\tau^2 = K\frac{V_{GM}^{\min}}{V_{GM}^{\max}}G_V|_{V_{GM}=V_{GM}^{\max}}\frac{\sqrt{tg^2(PM^*)+1}}{tg^2(PM^*)}\tau^2 = 1. \tag{30}$$

Combining (27) with (30), there is

$$\tau = \frac{\sqrt{2}}{8\alpha_{\min}\omega_{Gn}^{50}}\frac{\sqrt{1+\sqrt{1+4\left(2\cdot THD^*\frac{V_{GM}^{\min}}{V_{GM}^{\max}}\frac{\sqrt{tg^2(PM^*)+1}}{tg^2(PM^*)}\right)^2}}}{THD^*\frac{V_{GM}^{\min}}{V_{GM}^{\max}}\frac{\sqrt{tg^2(PM^*)+1}}{tg^2(PM^*)}}. \tag{31}$$

Consequently cf. (30),

$$K = \frac{2C_{DC}V_{DC}^*}{V_{GM}^{\min}\tau^2}\frac{tg^2(PM^*)}{\sqrt{tg^2(PM^*)+1}} \tag{32}$$

Considering typical values of $PM^* = \pi/4$ rad, $THD^* = 0.05$ and $\alpha_{\min} = 0.99$, the attainable crossover frequency range of DC link voltage loop with PI regulator is given by (cf. (29))

$$\frac{tg(PM^*)}{\tau} \leqslant \omega_C \leqslant \frac{KG_V|_{V_{GM}=V_{GM}^{\max}}}{\sqrt{2}}\tau\sqrt{1+\sqrt{1+\frac{4}{\left(KG_V|_{V_{GM}=V_{GM}^{\max}}\right)^2\tau^4}}}. \tag{33}$$

Considering (3) with universal range of magnitudes, there is

$$\omega_{C\min} = 2\pi\cdot 4.5 rad/s \leqslant \omega_C \leqslant 2\pi\cdot 10.7 rad/s = \omega_{C\max}. \tag{34}$$

It may then be concluded that the maximum attainable DC link voltage loop bandwidth is 10.7 Hz. Corresponding loop gains Bode diagram is depicted in Fig. 3, demonstrating the 238% uncertainty range of the crossover frequency values. It is evident that minimum phase margin and maximum total harmonic distortion (reflected by (24), see the green encirclement) values are also well-respected.

3. DOB-based DC link voltage bandwidth increase and uncertainty reduction

Typical systems employ a PI controller designed according to (32), attaining DC link voltage loop bandwidth given by (33). In order to increase the attainable bandwidth without deteriorating the grid side current THD, an additional degree-of-freedom is required so that the constraint (23) is not violated. DOB-based control [26–28] introduces the desired degree-of-freedom in one of the two following ways:

- By modifying the controller without affecting the existing plant, as shown in Fig. 4(top) [21,29–31];
- By modifying the plant without affecting the existing controller, as shown in Fig. 4(bottom) [32,33].

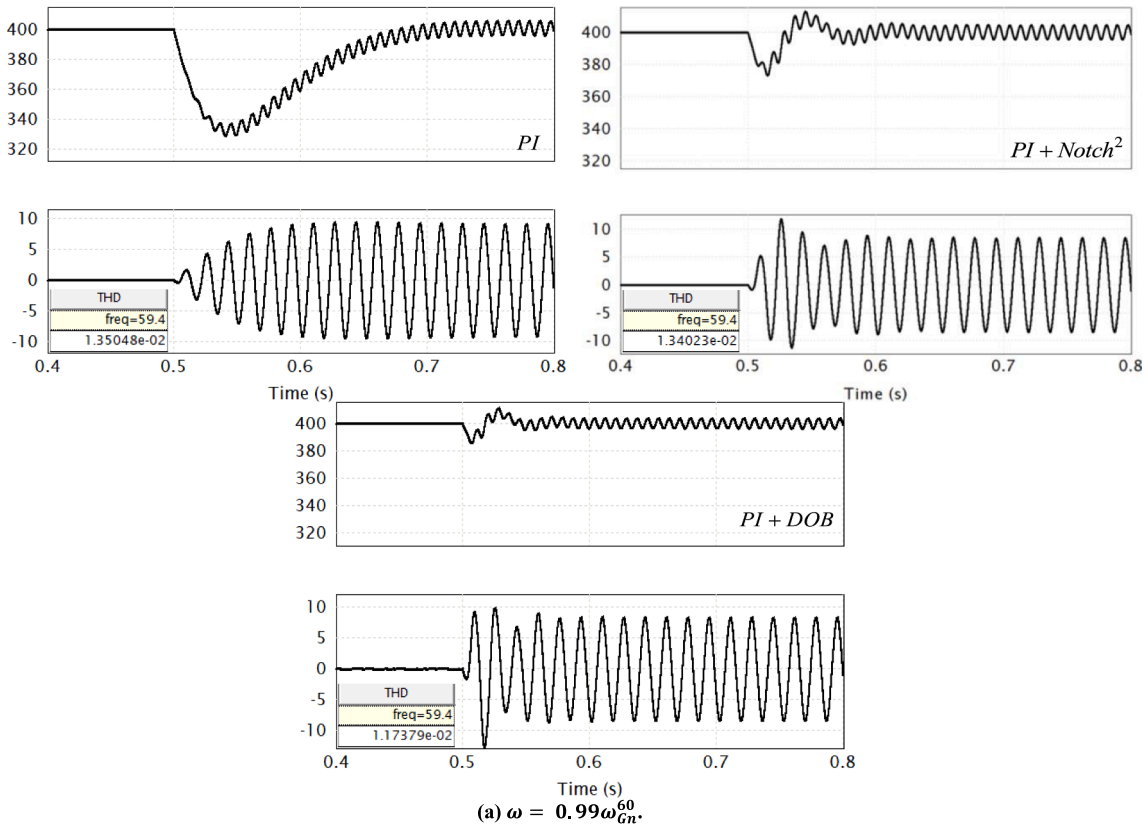
Note that in both cases the DOB is added to the system as shown in Fig. 5 yet the design is different for the two above-mentioned ways of operation. In this paper, the second way of operation is utilized so that modification of the existing controller is unnecessary and the DOB may be employed in plug-in fashion.

3.1. Defining the desired modified plant

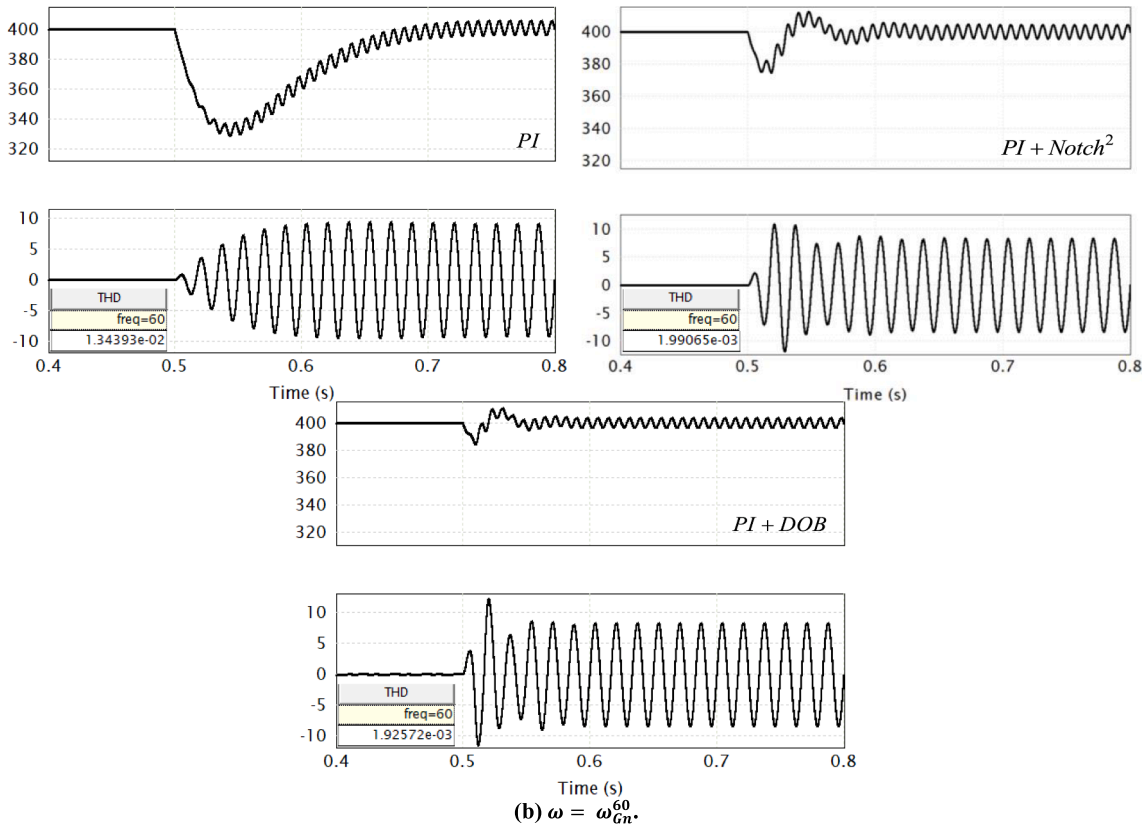
Referring to Fig. 5, modified plant ("seen" by existing PI controller) transfer function is defined as

$$P_V^{DOB}(s) \triangleq \frac{\widetilde{v}_{DC}(s)}{\widetilde{i}_{GMn}^*(s)}, \tag{35}$$

where i_{GMn}^* is the existing PI controller output. In order to increase the attainable DC link voltage loop bandwidth without violating (24), it is proposed to bring the modified plant to the following nominal form,

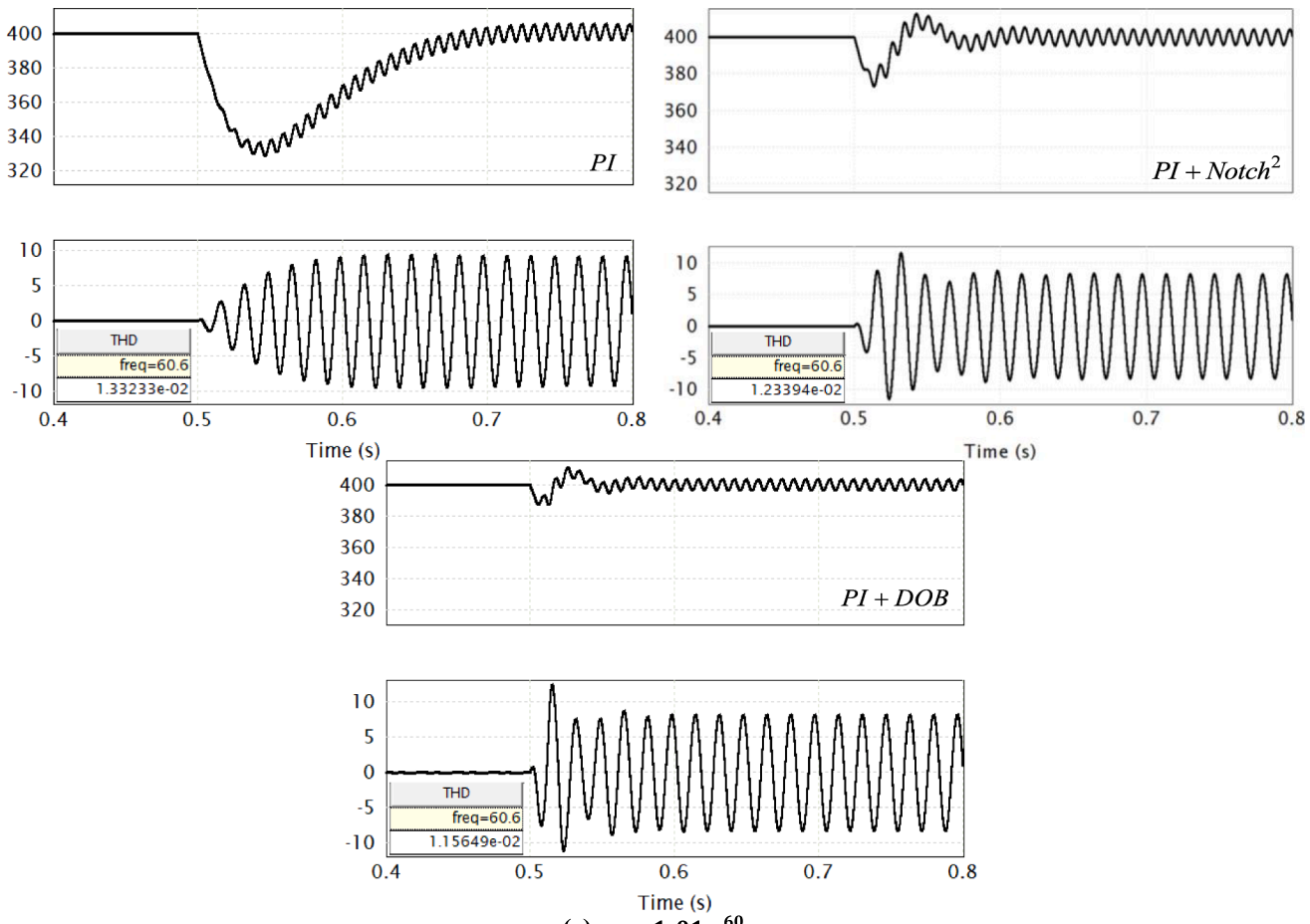


(a) $\omega = 0.99\omega_{Gn}^{60}$.



(b) $\omega = \omega_{Gn}^{60}$.

Fig. 9. Simulation results: DC link voltage (top) and grid-side current (bottom) responses to 0-to-500 W load step around ω_{Gn}^{60} for $V_{GM} = 85\sqrt{2}V$.



(c) $\omega = 1.01\omega_{Gn}^{60}$.

Fig. 9. (*continued*).

$$P_V^{DOB}(s) = \frac{G_V^{DOB}}{s} \frac{s^2 + \left(2\omega_{Gn}^{50}\right)^2}{s^2 + 4\xi\omega_{Gn}^{50}s + \left(2\omega_{Gn}^{50}\right)^2} \frac{s^2 + \left(2\omega_{Gn}^{60}\right)^2}{s^2 + 4\xi\omega_{Gn}^{60}s + \left(2\omega_{Gn}^{60}\right)^2} \tag{36}$$

with constant G_V^{DOB} satisfying

$$G_V^{DOB} = \delta \frac{V_{GM}^{\max}}{2C_{DC}V_{DC}^*} > G_V|_{V_{GM}=V_{GM}^{\max}} \Rightarrow \delta > 1 \tag{37}$$

to attain DC link voltage loop increase and

$$\left|P_V^{DOB}(2\alpha_{\min}\omega_{Gn}^{50})\right| \leqslant \frac{2 \cdot THD^*}{\left|C_V(2\alpha_{\min}\omega_{Gn}^{50})\right|} \tag{38}$$

to fulfil (21) and (24). Note that while the dual-frequency resonant term may be realized by placing two notch filters in series with the existing PI controller, constant utility voltage magnitude-independent gain cannot be attain using feedforward techniques. Consequently, (36) is attained by means of DOB as follows.

3.2. Attaining the desired modified plant

Modified plant dynamics may be formulated as

$$\begin{aligned}\widetilde{v}_{DC}(s) &= P_V^{DOB}(s)\left(\widetilde{i}_{GM}^*(s) + \widetilde{i}_{LUD}(s)\right) \\ &= P_V^{DOB}(s)\left(\widetilde{i}_{GMn}^*(s) + \widetilde{i}_{DOB}(s) + \widetilde{i}_{LUD}(s)\right)\end{aligned} \tag{39}$$

with i_{LUD} denoting corresponding lumped uncertainty and disturbance [34,35]. In case DOB in Fig. 5 accurately estimates i_{LUD}, its output is given by

$$\widetilde{i}_{DOB}(s) \approx -\widetilde{i}_{LUD}(s) \tag{40}$$

and then

$$\widetilde{v}_{DC}(s) = P_V^{DOB}(s)\widetilde{i}_{GMn}^*(s), \tag{41}$$

as desired. Lumped uncertainty and disturbance may obtained from (39) as

$$\widetilde{i}_{LUD}(s) = \widetilde{v}_{DC}(s)\left(P_V^{DOB}(s)\right)^{-1} - \widetilde{i}_{GM}^*(s) \tag{42}$$

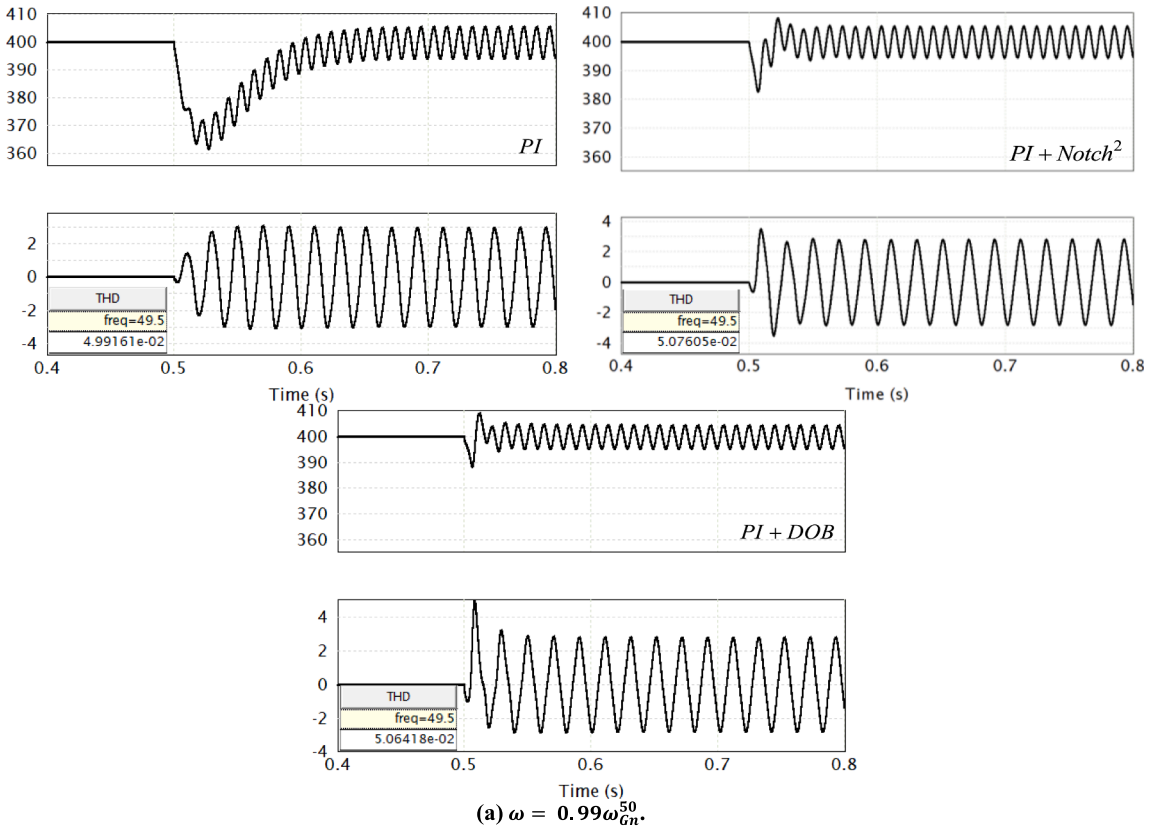
but cannot be utilized in (40) as is due to non-causality of (42) and the fact that modified plant inverse is improper. Instead, DOB methodology utilizes a filtered version of (42), namely [36,37]

$$\begin{aligned}\widetilde{i}_{DOB}(s) &= -\widetilde{i}_{LUD}(s)Q(s) = -\left(\widetilde{v}_{DC}(s)\left(P_V^{DOB}(s)\right)^{-1} - \widetilde{i}_{GM}^*(s)\right)Q(s) \\ &= \widetilde{i}_{GM}^*(s)Q(s) - \widetilde{v}_{DC}(s)\left(P_V^{DOB}(s)\right)^{-1}Q(s)\end{aligned} \tag{43}$$

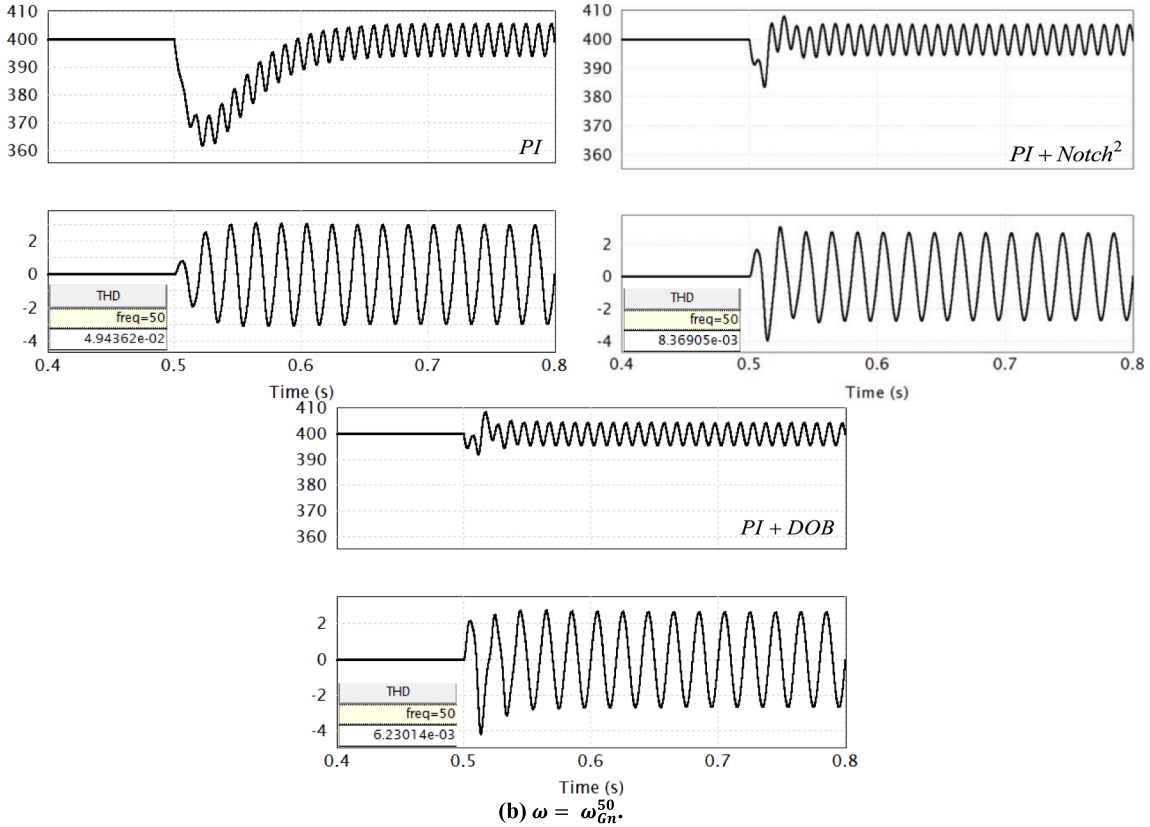
with $Q(s)$ denoting the transfer function of a linear filter so that $(P_V^{DOB}(s))^{-1} \bullet Q(s)$ is at least proper [38,39]. Combining (39) with (43) yields

$$\begin{aligned}\widetilde{v}_{DC}(s) &= P_V^{DOB}(s)\left(\widetilde{i}_{GMn}^*(s) - \widetilde{i}_{LUD}(s)Q(s) + \widetilde{i}_{LUD}(s)\right) \\ &= P_V^{DOB}(s)\left(\widetilde{i}_{GMn}^*(s) + \widetilde{i}_{LUD}(s)(1 - Q(s))\right).\end{aligned} \tag{44}$$

Consequently, the passband of $Q(s)$ should contain as much lumped

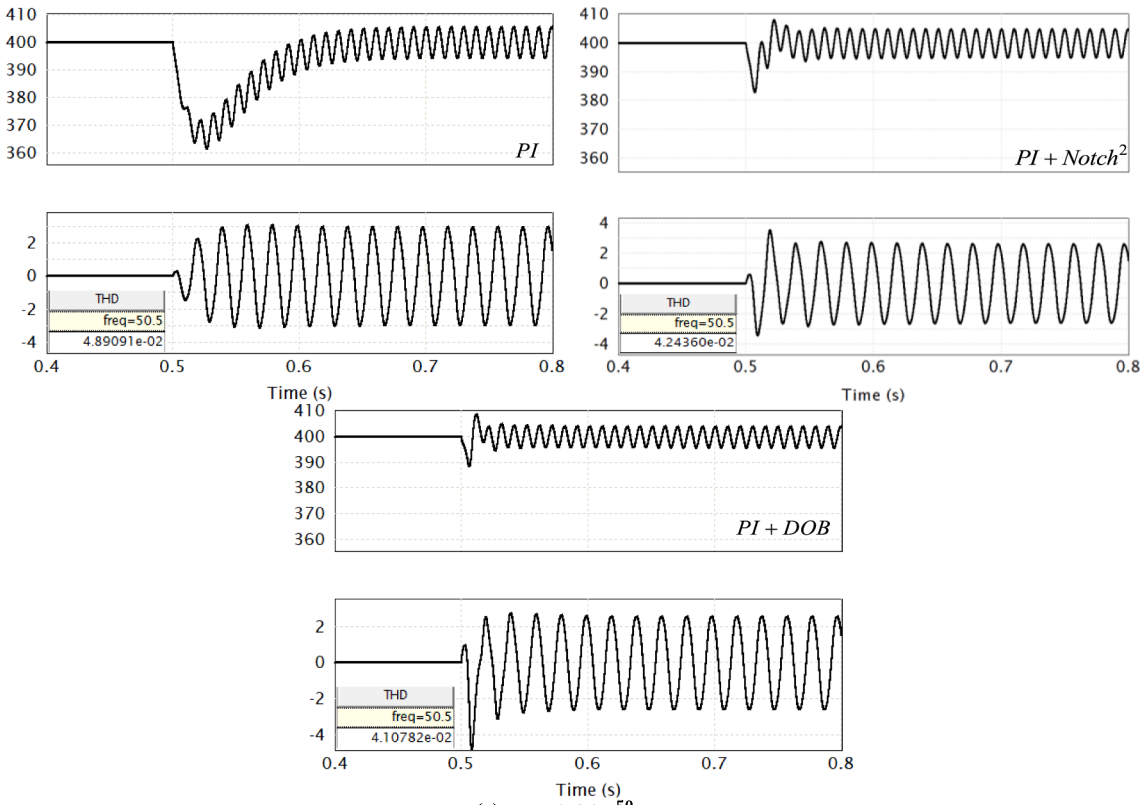


(a) $\omega = 0.99\omega_{Gn}^{50}$.



(b) $\omega = \omega_{Gn}^{50}$.

Fig. 10. Simulation results: DC link voltage (top) and grid-side current (bottom) responses to 0-to-500 W load step around ω_{Gn}^{50} for $V_{GM} = 265\sqrt{2}V$.



(c) $\omega = 1.01\omega_{Gn}^{50}$.

Fig. 10. (*continued*).

uncertainty and disturbance energy as possible in order to impose $1 - Q(s) \to 0$, allowing attaining (41). Selecting $Q(s)$ as the first-order low-pass filter

$$Q(s) = \frac{1}{\omega_Q^{-1}s+1} \Rightarrow \left(P_V^{DOB}(s)\right)^{-1} Q(s) = \frac{s}{G_V^{DOB}\left(\omega_Q^{-1}s+1\right)} \frac{s^2+4\xi\omega_{Gn}^{50}s+\left(2\omega_{Gn}^{50}\right)^2}{s^2+\left(2\omega_{Gn}^{50}\right)^2} \frac{s^2+4\xi\omega_{Gn}^{60}s+\left(2\omega_{Gn}^{60}\right)^2}{s^2+\left(2\omega_{Gn}^{60}\right)^2} \; is proper, \tag{45}$$

the DOB in Fig. 5 is realized as (cf. (43))

$$\widetilde{i}_{DOB}(s) = \widetilde{i}_{GM}^{*}(s)\frac{1}{\omega_Q^{-1}s+1} - \widetilde{v}_{DC}(s)\frac{s}{G_V^{DOB}\left(\omega_Q^{-1}s+1\right)} \frac{s^2+4\xi\omega_{Gn}^{50}s+\left(2\omega_{Gn}^{50}\right)^2}{s^2+\left(2\omega_{Gn}^{50}\right)^2} \frac{s^2+4\xi\omega_{Gn}^{60}s+\left(2\omega_{Gn}^{60}\right)^2}{s^2+\left(2\omega_{Gn}^{60}\right)^2}, \tag{46}$$

as shown in Fig. 6.

Using (46) with Fig. 5 and linearized plant in Fig. 2 yields

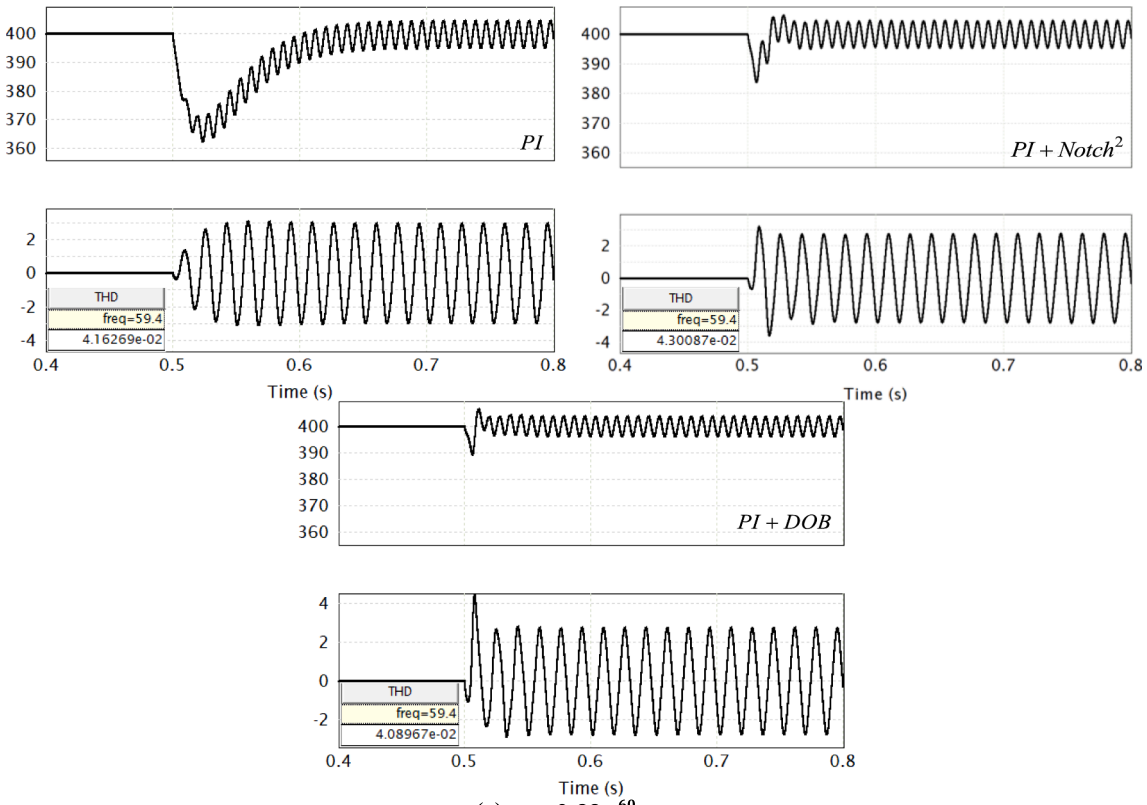
$$\widetilde{v}_{DC}(s) = \frac{P_V(s)\frac{1}{1-Q(s)}}{1+\frac{P_V(s)}{P_V^{DOB}(s)}\frac{Q(s)}{1-Q(s)}} \widetilde{i}_{GMn}^{*}(s) = \frac{P_V(s)}{1-Q(s)+\frac{P_V(s)}{P_V^{DOB}(s)}Q(s)} \widetilde{i}_{GMn}^{*}(s) \overset{Q(s)\approx 1}{=} P_V^{DOB}(s)\widetilde{i}_{GMn}^{*}(s). \tag{47}$$

Hence, selecting DOB filter cut-off frequency to reside well-above the highest frequency of interest yields (41) within the passband of $Q(s)$.

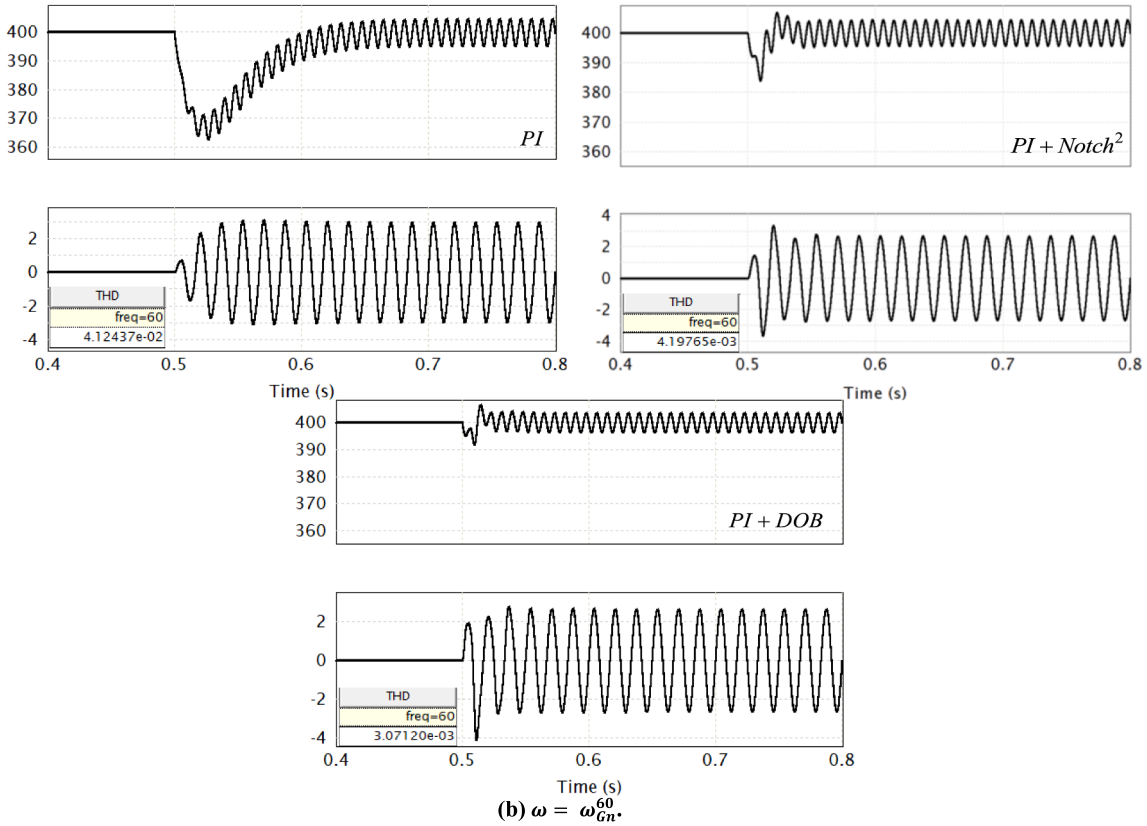
3.3. Selecting the desired modified plant parameters

Two modified plant parameters need to be defined, namely δ (cf. (37), which should be maximized to attain the highest possible DC link bandwidth and ξ, which should be high enough to allow sufficient attenuation at $\omega = 2\alpha_{min} \bullet \omega_G^{50}$), i.e. the two parameters are in obvious trade-off. Modified loop gain is

$$L_V^{DOB}(s) \triangleq C_V(s)P_V^{DOB}(s) \cong KG_V^{DOB}\frac{\tau s+1}{s^2} \frac{s^2+\left(2\omega_{Gn}^{50}\right)^2}{s^2+4\xi\omega_{Gn}^{50}s+\left(2\omega_{Gn}^{50}\right)^2} \frac{s^2+\left(2\omega_{Gn}^{60}\right)^2}{s^2+4\xi\omega_{Gn}^{60}s+\left(2\omega_{Gn}^{60}\right)^2}. \tag{48}$$

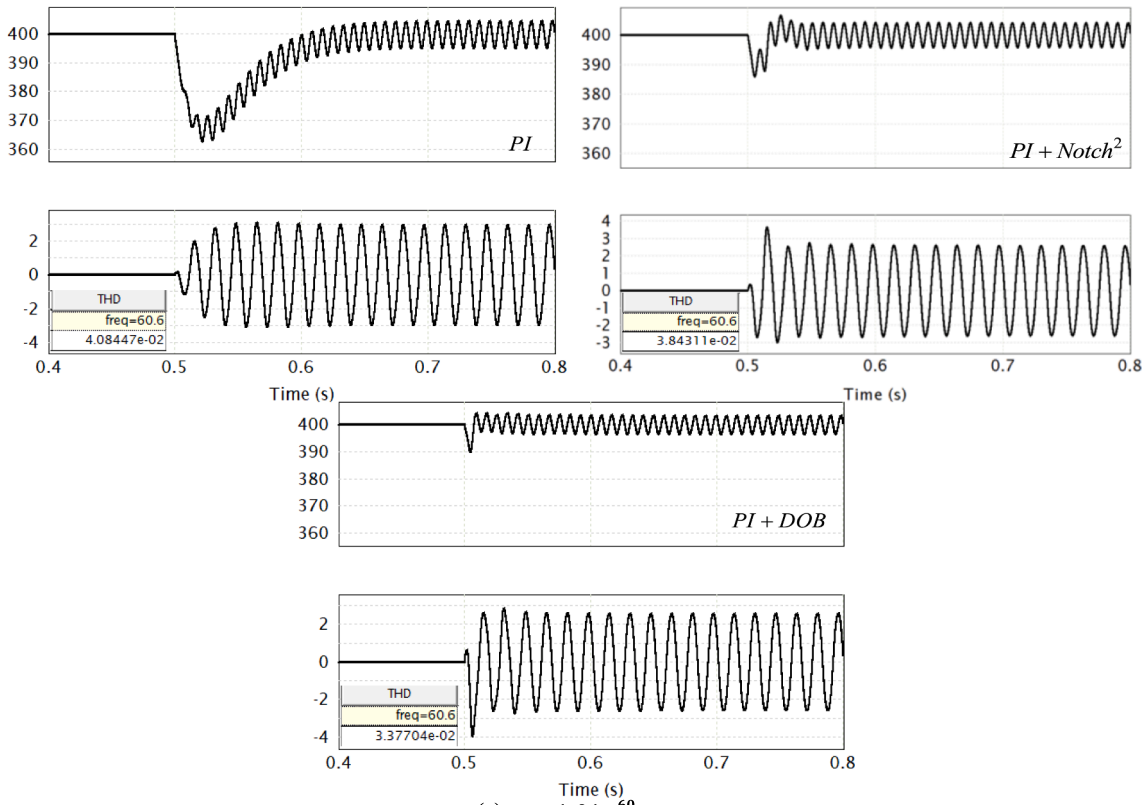


(a) $\omega = 0.99\omega_{Gn}^{60}$.



(b) $\omega = \omega_{Gn}^{60}$.

Fig. 11. Simulation results: DC link voltage (top) and grid-side current (bottom) responses to 0-to-500 W load step around ω_{Gn}^{60} for $V_{GM} = 265\sqrt{2}V$.



(c) $\omega = 1.01\omega_{Gn}^{60}$.

Fig. 11. (*continued*).

At the modified crossover frequency

$$\omega_C^{DOB} \triangleq \omega_{\left|L_V^{DOB}(\omega)\right|=1} \tag{49}$$

the following holds (temporarily neglecting the influence of both notch terms),

$$\left|L_V^{DOB}(\omega_C^{DOB})\right| = KG_V^{DOB}\frac{\sqrt{\tau^2(\omega_C^{DOB})^2+1}}{(\omega_C^{DOB})^2} = 1 \Rightarrow G_V^{DOB} = \frac{(\omega_C^{DOB})^2}{K\sqrt{\tau^2(\omega_C^{DOB})^2+1}}. \tag{50}$$

At the same time,

$$KG_V^{DOB}\frac{\sqrt{\tau^2(2\alpha_{\min}\omega_{Gn}^{50})^2+1}}{(2\alpha_{\min}\omega_{Gn}^{50})^2}\frac{1}{\sqrt{1+4\frac{\left(\alpha_{\min}(\omega_{Gn}^{50})^2\right)^2}{\left((\omega_{Gn}^{50})^2-(\alpha_{\min}\omega_{Gn}^{50})^2\right)^2}\xi^2}}\frac{1}{\sqrt{1+4\frac{(\alpha_{\min}\omega_{Gn}^{60}\omega_{Gn}^{50})^2}{\left((\omega_{Gn}^{60})^2-(\alpha_{\min}\omega_{Gn}^{50})^2\right)^2}\xi^2}} \leqslant 2\cdot THD^* \tag{51}$$

must hold in order to satisfy (38). Considering the boundary case of (51) and rearranging yields

$$KG_V^{DOB} = K\delta G_V|_{V_{GM}=V_{GM}^{\max}} = \frac{8\cdot THD^*\cdot(\alpha_{\min}\omega_{Gn}^{50})^2}{\sqrt{4\tau^2(\alpha_{\min}\omega_{Gn}^{50})^2+1}}\sqrt{1+4\frac{(\alpha_{\min}\omega_{Gn}^{50}\omega_{Gn}^{50})^2}{\left((\omega_{Gn}^{50})^2-(\alpha_{\min}\omega_{Gn}^{50})^2\right)^2}\xi^2}\sqrt{1+4\frac{(\alpha_{\min}\omega_{Gn}^{60}\omega_{Gn}^{50})^2}{\left((\omega_{Gn}^{60})^2-(\alpha_{\min}\omega_{Gn}^{50})^2\right)^2}\xi^2}. \tag{52}$$

Consequently (cf. (29)),

$$\delta = \sqrt{1+4\frac{(\alpha_{\min}\omega_{Gn}^{50}\omega_{Gn}^{50})^2}{\left((\omega_{Gn}^{50})^2-(\alpha_{\min}\omega_{Gn}^{50})^2\right)^2}\xi^2}\sqrt{1+4\frac{(\alpha_{\min}\omega_{Gn}^{60}\omega_{Gn}^{50})^2}{\left((\omega_{Gn}^{60})^2-(\alpha_{\min}\omega_{Gn}^{50})^2\right)^2}\xi^2}. \tag{53}$$

Considering $\alpha_{\min} = 0.99$, (53) reduces to

$$\delta = \sqrt{1+9.9\cdot10^3\xi^2}\sqrt{1+26.7\xi^2}. \tag{54}$$

As shown in [16,25], crossover frequency of a loop containing a notch filter should be kept below half-notch frequency to avoid notch-frequency oscillations during transients. Consequently, modified crossover frequency of

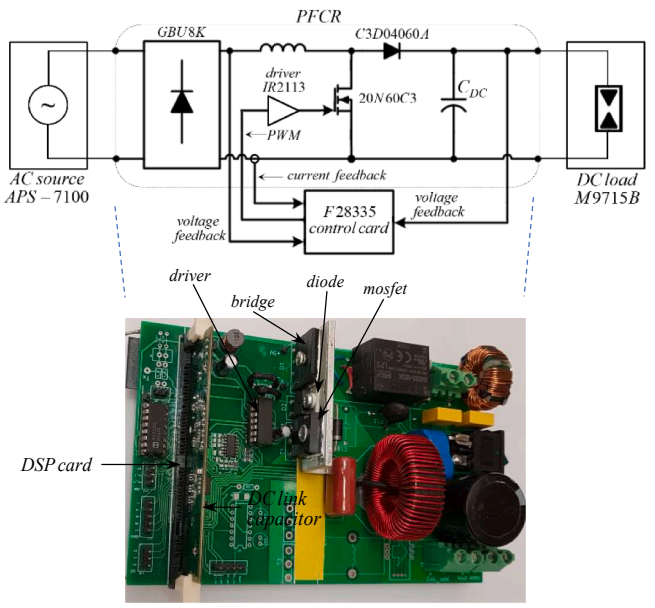


Fig. 12. Experimental setup schematic.

$$\omega_C^{DOB} = \omega_{Gn}^{50} \tag{55}$$

is suggested in this paper. Considering values of $PM^* = \pi/4$ rad, $THD^* = 0.05$ while substituting (55) into (50) and then into (52) and (54) yields

$$\delta \cong 5, \quad \xi \cong 0.05. \tag{56}$$

Substituting $\delta = 5$ into (54) and solving yields $\xi = 0.05$. Taking into account (47), actual loop gain is given by

$$L_V^{DOB}(s) = C_V(s)\frac{P_V(s)}{1 - Q(s) + \frac{P_V(s)}{P_V^{DOB}(s)}Q(s)} == C_V(s)P_V^{DOB}\frac{P_V(s)}{P_V^{DOB}(1 - Q(s)) + P_V(s)Q(s)}$$

$$= \underbrace{C_V(s)P_V(s)}_{L_V(s)}\frac{1}{1 - Q(s) + \frac{P_V(s)}{P_V^{DOB}(s)}Q(s)} = L_V(s)\underbrace{\frac{\omega_Q^{-1}s + 1}{\omega_Q^{-1}s + \frac{V_{GM}}{\delta V_{GM}^{\max}}\frac{s^2 + \left(2\omega_{Gn}^{50}\right)^2}{s^2 + 4\xi\omega_{Gn}^{50}s + \left(2\omega_{Gn}^{50}\right)^2}\frac{s^2 + \left(2\omega_{Gn}^{60}\right)^2}{s^2 + 4\xi\omega_{Gn}^{60}s + \left(2\omega_{Gn}^{60}\right)^2}}}_{G(s)} = L_V(s)G(s), \tag{57}$$

reducing to (41) for $Q(s) = 1$. Setting the DOB filter cut-off frequency one decade above the highest frequency of interest, namely

$$\omega_Q = 10\cdot\left(2\omega_{Gn}^{60}\right) \tag{58}$$

yields corresponding loop gains Bode diagram depicted in Fig. 7 (top). Corresponding crossover frequencies range is

$$2\pi\cdot 43 rad/s \leqslant \omega_C \leqslant 2\pi\cdot 49 rad/s, \tag{59}$$

demonstrating the 14% uncertainty range of the crossover frequency values, as opposed to 238% attained by classical PI control (cf. Fig. 3). Moreover, minimum phase margin value is well-respected. In order to verify that the maximum total harmonic distortion is also complied with, zoomed version of magnitude plots is depicted in Fig. 7(bottom). It may be concluded that (24) is respected within frequency range (4).

4. Validation

A 500 W-rated power factor correction front end (PFC) with $C_{DC} =$ 385μF (allowing hold-up time of 10 ms [10]), operated with $V_{DC}^* = 400V$ is considered for the proposed methodology validation. Adopting typical desired performance merits $THD^* = 0.05$ and $PM^* = 45^\circ$ with grid frequency uncertainty of 1%, (31) and (32) yield the following PI controller coefficients,

$$K = 1.44, \quad \tau = 0.0355\,. \tag{60}$$

In PI + DOB case, the observer was plugged-in according to (46) with (37), (56) and (58).

4.1. Simulations

Simulations were carried out by PSIM software. In order to further compare the proposed control structure with the state of the art, performance of PI + DOB controller is compared to that of PI + Notch^2 controller [40]

$$C_V(s) = K\frac{\tau s + 1}{s}\frac{s^2 + \left(2\omega_{Gn}^{50}\right)^2}{s^2 + 4\xi\omega_{Gn}^{50}s + \left(2\omega_{Gn}^{50}\right)^2}\frac{s^2 + \left(2\omega_{Gn}^{60}\right)^2}{s^2 + 4\xi\omega_{Gn}^{60}s + \left(2\omega_{Gn}^{60}\right)^2} \tag{61}$$

designed according to the above-mentioned performance merits pair. The following optimized coefficients were obtained for the controller (61),

$$K = 25.4, \quad \tau = 0.0043, \quad \xi = 0.019. \tag{62}$$

Time-domain simulation results depicting the responses of converters' DC link voltage and grid-side current to zero-to-rated load step for boundary values of V_{GM} and boundary and nominal values of ω_G under PI, PI + Notch^2 and PI + DOB control are given in Figs. 8 – 11. Corresponding values of grid-side current THD, calculated by simulation software in steady-state, are also indicated on respective subplots. It may be concluded that:

- Under PI control, grid frequency has nearly no influence on DC link voltage response. On the other hand, mains voltage magnitude increase rises the crossover frequency (cf. (34)), leading to shorter settling time. For $V_{GM} = 85\sqrt{2}$ V the 2% settling time is in a good agreement with corresponding bandwidth since$5/\omega_{Cmin} = 177$ ms. Likewise, for $V_{GM} = 265\sqrt{2}$ V the 2% settling time corresponds well with the bandwidth since$5/\omega_{Cmax} = 75$ ms. DC link voltage undershoot is ~20% for $V_{GM} = 85\sqrt{2}$ V and ~10% for $V_{GM} = 265\sqrt{2}$ V. Furthermore, the maximum value of THD is expected to be attained (cf. Fig. 3) under $V_{GM} = 265\sqrt{2}$ V at $\omega = \alpha_{min}\omega_{Gn}^{50}$, while the minimum value of THD is expected to occur under $V_{GM} = 85\sqrt{2}$ V at $\omega = \alpha_{max}\omega_{Gn}^{60}$. The two boundaries are accurately supported by simulation results with 1.33% and 4.99% minimum and maximum THD values, respectively.

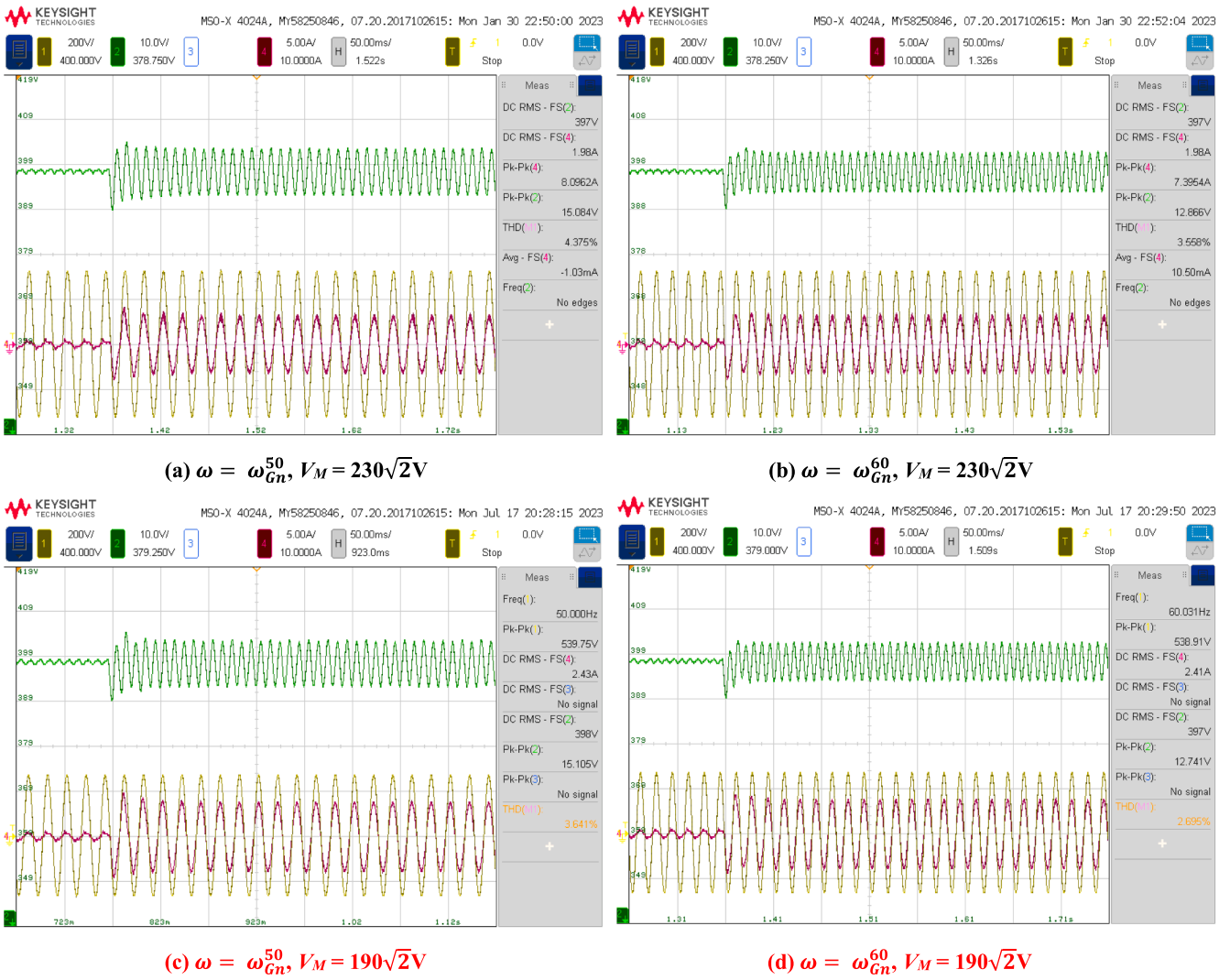


Fig. 13. Experimental results: Transient performance, PI + DOB controller.

- Under PI + Notch^2 control, grid frequency has nearly no influence on DC link voltage response. On the other hand, mains voltage magnitude increase rises the crossover frequency (as expected, since mains voltage magnitude variations are not compensated), leading to shorter settling time and reduced undershoot. Furthermore, the maximum value of THD is attained under $V_{GM} = 265\sqrt{2}$ V at $\omega = \alpha_{min}\omega_{Gn}^{50}$, while the minimum value of THD is attained under $V_{GM} = 85\sqrt{2}$ V at $\omega = \alpha_{max}\omega_{Gn}^{60}$.
- Under PI + DOB control, both grid frequency and mains voltage magnitude have nearly no influence on DC link voltage response, as desired. The 2% settling time is around half mains cycle irrespectively of main frequency and voltage magnitude, as expected from (59). DC link voltage undershoot is around $\sim$ 2.5%, again irrespectively of main frequency and voltage magnitude. Note that the dynamic performance attained by PI + DOB control is slightly better than its PI + Notch^2 counterpart. This is in good agreement with the results demonstrated in [21]. Furthermore, the maximum value of THD is expected to be attained (cf. Fig. 6) under $V_{GM} = 265\sqrt{2}$ V at $\omega = \alpha_{min}\omega_{Gn}^{50}$, while the minimum value of THD is expected to occur under $V_{GM} = 85\sqrt{2}$ V at $\omega = \omega_{Gn}^{60}$. The two boundaries are accurately supported by simulation results with 0.19% and 5% minimum and maximum THD values, respectively.

4.2. Experiments

For the purpose of validating the presented methodology, matching experimental setup shown in Fig. 12 was utilized with PFCR inductance of 315μH. The control algorithm [41] was realized adopting first-order hold digitization [42] by TMS320F28335 DSP-based control card [43] under average current controlled inner loop with 150 kHz switching/sampling frequency. The utility was emulated by APS-7100 Gwinstek programmable AC power source operating as 50 Hz/60 Hz ±1%, 230V_{rms} sinusoidal voltage source. M9715B Maynuo DC electronic load functioning in constant power mode was employ to emulate PFCR load p_L (cf. Fig. 1). It was shown in [32, Fig. 16] that APS-7100 Gwinstek power source generates AC voltage possessing up to 0.5% total harmonic distortion, hence THD of experimental current waveforms may contain correspondingly higher harmonic content. Moreover, due to hardware limitations performance for $230\sqrt{2}$ V and $190\sqrt{2}$ V utility magnitudes only was examined experimentally.

Experimental results depicting the responses of converters' DC link voltage and grid-side current to zero-to-rated load step for the two nominal values of ω_G under PI + DOB control are given in Fig. 13. It may be concluded that experimental results are in close agreement with corresponding simulation counterparts, validating the dynamic performance under proposed control. It is well-evident that the latter is nearly independent of operating frequency and mains magnitude value, as

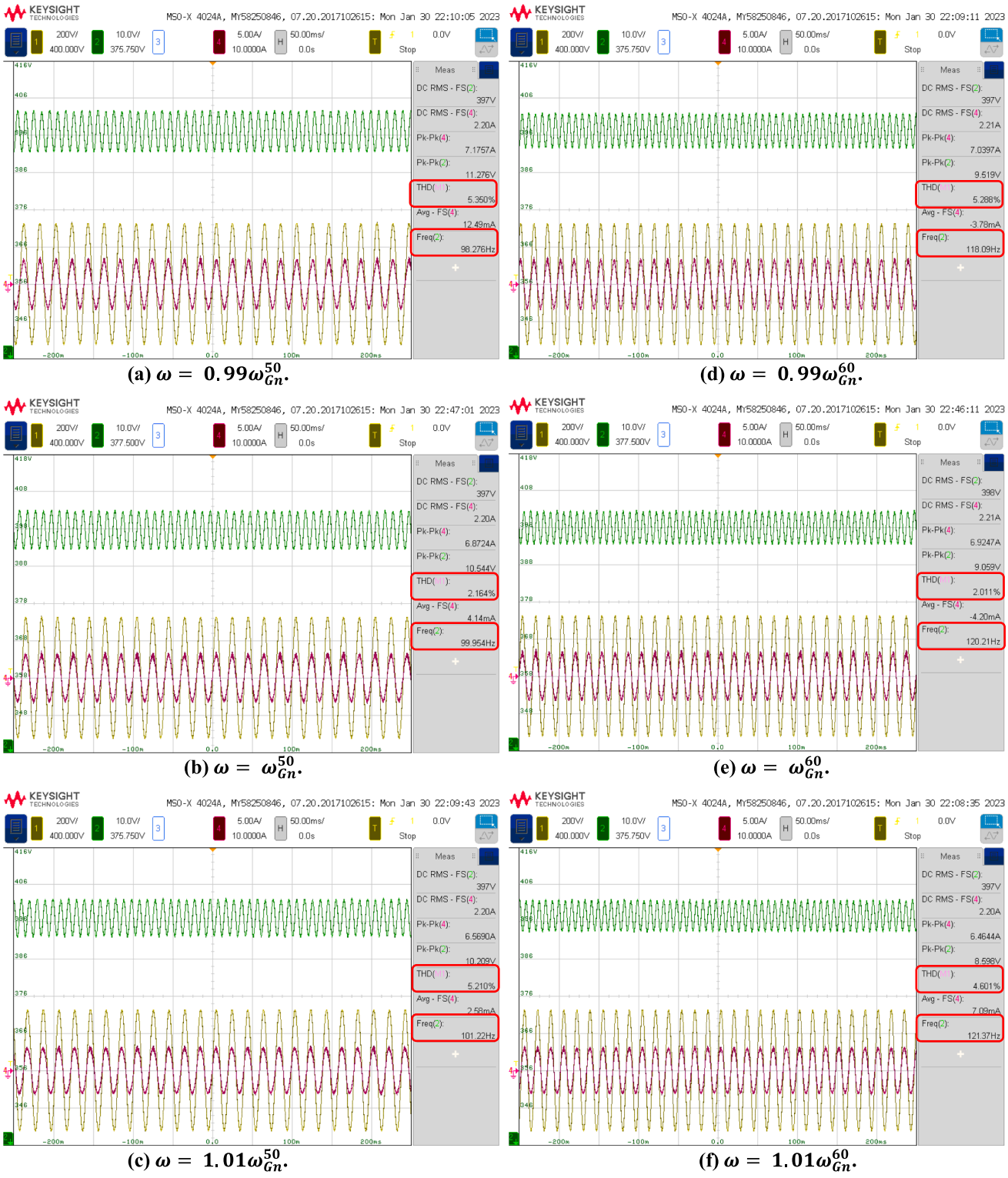


Fig. 14. Experimental results: Steady-state performance, $V_M = 230\sqrt{2}$ V, PI + DOB controller.

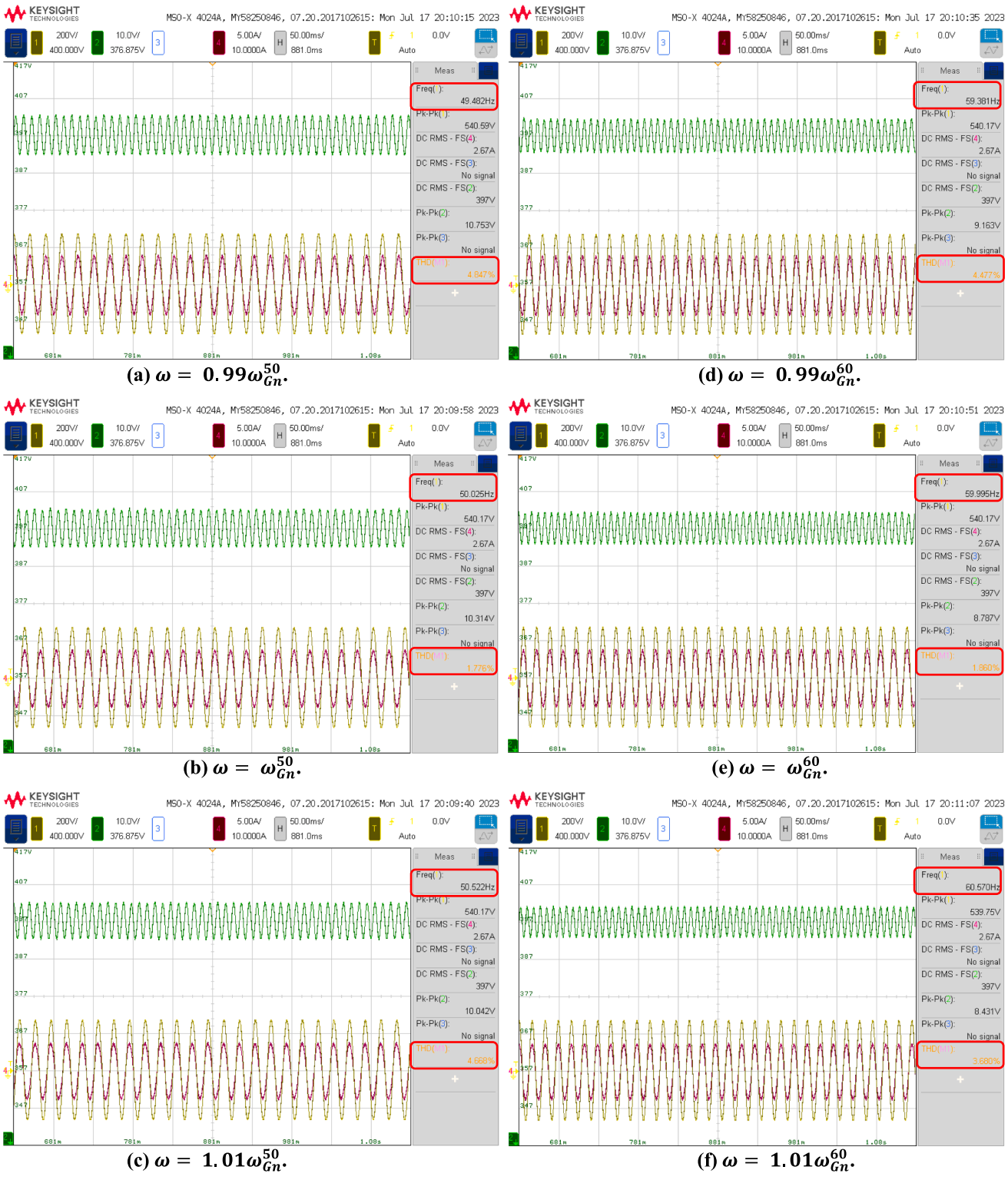


Fig. 15. Experimental results: Steady-state performance, $V_M = 190\sqrt{2}$ V, PI + DOB controller.

expected. Steady-state operational waveforms of DC link voltage and grid-side current for boundary and nominal values of ω_G under PI + DOB control are depicted in Figs. 14 and 15 for $230\sqrt{2}$ V and $190\sqrt{2}$ V utility magnitudes, respectively. Corresponding values of grid-side current THD are also indicated on oscilloscope screenshots. As predicted by simulations, the maximum value of THD is attained at $\omega = \alpha_{min}\omega_{Gn}^{50}$, while the minimum value of THD is attained at $\omega = \omega_{Gn}^{60}$.

5. Remark: Operation with distorted grid

In case of distorted mains, utility-side voltage contains odd harmonics with arbitrary phases, i.e.

$$v_G(t) \approx \sum_{n=1,odd}^{\infty} v_{GMn}(t)\sin\left(\int n\omega_G(t)dt + \theta_n\right). \tag{63}$$

Considering near-unity displacement factor operation, grid current

may be approximated by

$$i_G(t) \approx i_{GM}(t)\sin\left(\int \omega_G(t)dt + \theta_1\right). \tag{64}$$

respectively. Utility-side instantaneous power is then given by (assuming $\theta_1 = 0$ for brevity)

$$p_G(t) = v_G(t)i_G(t) = \underbrace{0.5v_{GM1}(t)i_{GM}(t)}_{p_{GM}(t)} - \underbrace{\underbrace{0.5v_{GM1}(t)i_{GM}(t)}_{p_{GM}(t)}\cos\left(2\int \omega_G(t)dt\right) + 0.5\sum_{n=3,odd}^{\infty} v_{GMn}(t)i_{GM}(t)\left(\cos\left(\int (n-1)\omega_G(t)dt + \theta_n\right) - \cos\left(\int (n+1)\omega_G(t)dt + \theta_n\right)\right)}_{\Delta p_{GM}(t)}. \tag{65}$$

In case the average value of $v_{DC}(t)$ is regulated to V_{DC}^* and mains base frequency is constant, (7) becomes

$$v_{DC}^{ss}(t) \approx V_{DC}^* - \Delta v_{DC}(t) \tag{66}$$

with

$$\Delta v_{DC}(t) \approx \frac{P_L}{2\omega_G C_{DC} V_{DC}^*}\sin(2\omega_G t) - \frac{I_{GM}}{2C_{DC}\omega_G V_{DC}^*}\sum_{n=3,odd}^{\infty} V_{GMn}\left(\frac{\sin((n-1)\omega_G t + \theta_n)}{n-1} - \frac{\sin((n+1)\omega_G t + \theta_n)}{n+1}\right) = \sum_{n=2,even}^{\infty} \Delta V_{DCn}\sin(n\omega_G t + \phi_n) \tag{67}$$

with

$$\Delta V_{DCn} << V_{DC}^* \tag{68}$$

in practice for even n. In terms of second-base-frequency harmonic component, the worst case arises when $\theta_3 = \pi$ so that

$$\Delta v_{DC}(t) \approx \underbrace{\frac{P_L + V_{GM3}I_{GM}}{2\omega_G C_{DC} V_{DC}^*}}_{\Delta V_{DC2}}\sin(2\omega_G t) + \sum_{n=4,even}^{\infty} \Delta V_{DCn}\sin(n\omega_G t + \phi_n). \tag{69}$$

It is well evident that ΔV_{DC2} in (69) is higher than ΔV_{DC} arising in harmonic-free mains voltage case (cf. (11)). Moreover, ripple voltage in (69) contains additional harmonics residing at even base-frequency multiples. Further considering 3rd and 5th mains voltage harmonics only for brevity with $\theta_5 = \theta_3$ as the worst case, (69) reduces to

$$\Delta v_{DC}(t) \approx \frac{(2P_L + I_{GM}V_{GM3})\sin(2\omega_G t) - 0.5(I_{GM}V_{GM3} + I_{GM}V_{GM5})\sin(4\omega_G t) - 0.33I_{GM}V_{GM5}\sin(6\omega_G t)}{4\omega_G C_{DC} V_{DC}^*}. \tag{70}$$

Using (66), (67) with Fig. 1(b) while applying (68) yields

$$i_G^{ss}(t) \approx \frac{2P_L}{V_{GM}16P_L\omega_G V_{DC}^* C_{DC}} \times \begin{pmatrix} \sin(\omega_G t) + \\ V_{GM}(2P_L + I_{GM}V_{GM3})|C_V(2\omega_G)|\sin(3\omega_G t + \arg C_V(2\omega_G)) + \\ 0.5V_{GM}(I_{GM}V_{GM3} + I_{GM}V_{GM5})|C_V(4\omega_G)|\sin(3\omega_G t + \arg C_V(4\omega_G)) - \\ 0.5V_{GM}(I_{GM}V_{GM3} + I_{GM}V_{GM5})|C_V(4\omega_G)|\sin(5\omega_G t + \arg C_V(4\omega_G)) + \\ 0.33V_{GM}I_{GM}V_{GM5}|C_V(6\omega_G)|\sin(5\omega_G t + \arg C_V(6\omega_G)) - \\ 0.33V_{GM}I_{GM}V_{GM5}|C_V(6\omega_G)|\sin(7\omega_G t + \arg C_V(6\omega_G)) \end{pmatrix} = \sum_{n=1,odd}^{7} i_{Gn}^{SS}\sin(n\omega_G t + \delta_n). \tag{71}$$

i.e. first, third, fifth and seventh harmonics dominate the utility-side current (without considering switching-frequency components). Corresponding utility-side current THD is bounded by

$$THD = \frac{\left|\sqrt{\left(i_{G3}^{ss}\right)^2 + \left(i_{G5}^{ss}\right)^2 + \left(i_{G7}^{ss}\right)^2}\right|}{i_{G1}^{ss}}. \quad (72)$$

Since the third harmonic magnitude in (71) is obviously higher than its counterpart in (12), the value of (72) would be higher than that of (13). Moreover, in order to keep (72) below certain limit, controller gain magnitude should also be bounded (cf. (21)) at $4\omega_G$ and $6\omega_G$, calling for additional notching pairs at corresponding frequencies. Design of the controller based on proposed methodology for distorted mains case is left as the future work.

6. Conclusion

In this paper, design of a novel dual-frequency-tuned utility magnitude insensitive plug-in disturbance observer (DOB), supporting universal utility frequencies and magnitudes was proposed, allowing to enhance the dynamic response of grid-connected converters equipped with PI or type-II DC link voltage controller without altering the coefficients of the latter. The improved performance is attained by robust modification of DC link voltage loop plant while estimating and cancelling grid voltage magnitude variations. Comprehensive PI and DOB controller design guidelines are provided for desired values of utility-side current total harmonic distortion and DC link voltage loop phase margin Simulations and experimental results validate the validity of proposed method.

CRediT authorship contribution statement

Yan Vule: Investigation, Methodology, Validation. **Alon Kuperman:** Conceptualization, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

[1] Zha X, Huang M, Liu Y, Tian Z. An overview on safe operation of grid-connected converters from resilience perspective: Analysis and design. Int. J. Electr. Power Energy Syst. Dec. 2022;143:108511.

[2] Taul MG, et al. Current reference generation based on next-generation grid code requirements of grid-tied converters during asymmetrical faults. IEEE J Emerg Sel Top Power Electron Dec. 2020;8(4):3784–97.

[3] Castellan S, Buja G, Menis R. Single-phase power line conditioning with unity power factor under distorted utility voltage. Int. J. Electr. Power Energy Syst. Oct. 2020;121:106057.

[4] Mellincovsky M, et al. Low-frequency DC-link ripple elimination in power converters with reduced capacitance by multiresonant direct voltage regulation. IEEE Trans. Ind. Electron. Mar. 2017;64(3):2015–23.

[5] Strajnikov P, Peretz MM, Kuperman A. Reduced-rating-electronic-capacitor-based DC links for power electronics supported distributed energy resources. IEEE Trans. Smart Grid Sep. 2022;13(5):3943–53.

[6] Khajehoddin SA, Karimi-Ghartemani M, Jain PK, Bakhshari A. DC-bus design and control for a single-phase grid-connected renewable converter with a small energy storage component. IEEE Trans. Power Electron. Jul. 2013;28(7):3245–54.

[7] Mallik A, Lu J, Khaligh A. A comparative study between PI and type-II compensators for H-bridge PFC converter. IEEE Trans. Ind. Appl. Mar. 2018;54(2):1128–35.

[8] Nabatirad M, Razzaghi R, Bahrani B. Autonomous power balance in hybrid AC/DC microgrids. Int. J. Electr. Power Energy Syst. Mar. 2023;146:108752.

[9] Liu B, et al. A modified modulation for single-phase photovoltaic/battery inverter to improve AC power quality with low DC link capacitance. Int. J. Electr. Power Energy Syst. Sept. 2019;110:705–12.

[10] Strajnikov P, Kuperman A. Guidelines for voltage controller coefficients design to attain prescribed grid current THD and DC link voltage undershoot values in power factor correction front ends. IEEE J Emerg Sel Top Power Electron Dec. 2022;10(6):6523–33.

[11] A. Kuperman, On the explicit relation between mains-side current THD and DC link voltage controller gain in grid-connected converters, in Proc. IEEE Int. Conf. Intel. Eng. Syst., Aug. 12 – 15, Crete, Greece, 2022.

[12] Sebastian J, et al. On the maximum bandwidth attainable by power factor correctors with a standard compensator. IEEE Trans. Ind. Appl. Aug. 2010;46(4):1485–97.

[13] Merai M, Naouar MW, Slama-Belkhodja I, Mohnmasson E. A systematic design methodology for DC-link voltage control of single-phase grid-tied PV systems. Math Comp Simul May 2021;183:158–70.

[14] Karimi-Ghartemani M, Khajehoddin SA, Jain PK, Bakhshari A. A systematic approach to DC-bus control design in single-phase grid-connected renewable converters. IEEE Trans. Power Electron. Jul. 2013;28(7):3158–66.

[15] El Aroudi A, Cid-Pastor A, Martinez-Salamero L. Suppression of line frequency instabilities in PFC AC-DC power supplies by feedback notch filtering the pre-regulator output voltage. IEEE Trans Circ Syst – I: Reg Papers, Mar 2013;60(3):796–809.

[16] Strajnikov P, Kuperman A. "Selection of PI+Notch voltage controller coefficients to attain desired steady-state and transient performance in PFC rectifiers," *IEEE. J Emerg Sel Top Power Electron Dec. 2022;10(6):6534–44.*

[17] Taghizadeh S, Hossain MJ, Lu J, Karimi-Ghartemani M. An enhanced DC-bus voltage control loop for single-phase grid-connected DC/AC converters. IEEE Trans. Power Electron. Jun. 2019;34(6):5819–29.

[18] Pannawan A, et al. Fast bus voltage control of single-phase grid-connected converter with unified harmonic mitigation. IEEE Access 2023;11:6452–66.

[19] Somkun S, et al. Adaptive notch filters for bus voltage control and capacitance degradation prognostic of single-phase grid-connected inverter. IEEE Trans. Ind. Electron. Dec. 2023;70(12):12190–200.

[20] Cho Y, Lai J-S. Digital plug-in repetitive controller for single-phase bridgeless PFC converters. IEEE Trans. Ind. Electron. Jan. 2013;28(1):165–75.

[21] Vule Y, Kuperman A. Plug-in disturbance observer assisted DC link voltage control of grid-connected converters to improve transient performance without deteriorating grid current quality. Int. J. Electr. Power Energy Syst. Dec. 2022;143:108439.

[22] Kim H, Shim H, Jo NH. Adaptive add-on output regulator for rejection of sinusoidal disturbances and application to optical disc drives. IEEE Trans. Ind. Electron. Oct. 2014;61(10):5490–9.

[23] Erickson RW, Maksimovic D. Fundamentals of Power Electronics. Kluwer Academic; 2001.

[24] Strajnikov P, Kuperman A. On the minimum DC link capacitance in practical PFC rectifiers considering THD requirements and load transients. IEEE Trans. Ind. Electron. Nov. 2022;69(11):11067–75.

[25] Strajnikov P, Kuperman A. DC-link capacitance reduction in PFC rectifiers employing PI+Notch voltage controllers. IEEE Trans. Power Electron. Jan. 2023;38(1):977–86.

[26] Chen W-H, Ohnishi K, Guo L. Advances in disturbance/uncertainty estimation and attenuation. IEEE Trans. Ind. Electron. Sep. 2015;62(9):5758–62.

[27] Chen W-H, Yang J, Guo L, Li S. Disturbance-observer-based control and related methods—An overview. IEEE Trans. Ind. Electron. Feb. 2016;63(2):1083–95.

[28] Yang J, Chen W-H, Ding Z. Disturbance observers and applications. Trans. Inst. Meas. Control 2016;38(6):621–4.

[29] Alshek N, et al. Modified uncertainty and disturbance estimator for enhanced periodic signals suppression. IEEE Trans. Ind. Electron. Feb. 2019;66(2):1246–54.

[30] Ali M, Yaqoob M, Cao L, Loo K-H. Disturbance-observer-based DC-bus voltage control for ripple mitigation and improved dynamic response in two-stage single-phase inverter system. IEEE Trans. Ind. Electron. Sep. 2019;66(9):6836–45.

[31] Dai C, Guo T, Yang J, Li S. A disturbance observer-based current –constrained controller for speed regulation of PMSM system subject to unmatched disturbances. IEEE Trans. Ind. Electron. Jan. 2021;68(1):767–75.

[32] Sitbon M, Schacham S, Kuperman A. Disturbance observer-based voltage regulation of current-mode-boost-converter-interfaced photovoltaic generator. IEEE Trans. Ind. Electron. Sep. 2015;62(9):5776–85.

[33] Vule Y, Elkayam M, Kuperman A. Uncertainty and disturbance estimator with improved steady-state performance for grid-connected power converters. IET Renew Power Gen 2020;14(12):2183–91.

[34] Gadelovits S, et al. UDE-based controller equipped with a multi-band-stop filter to improve the voltage quality of inverters. IEEE Trans. Ind. Electron. Sep. 2017;64(9):7433–43.

[35] Gadelovits S, et al. Uncertainty and disturbance estimator-based controller equipped with a multiple-time-delayed filter to improve the voltage quality of inverters. IEEE Trans. Ind. Electron. Nov. 2019;66(11):8947–57.

[36] Sariyildiz E, Ohnishi K. Stability and robustness of disturbance-observer-based motion control systems. IEEE Trans. Ind. Electron. Jan. 2015;62(1):414–22.

[37] Gadelovits S, et al. Uncertainty and disturbance estimator-based controller equipped with a time-delayed filter to improve the voltage quality of inverters. IEEE Trans. Ind. Electron. Jan. 2019;66(1):459–69.

[38] Aharon I, Shmilovitz D, Kuperman A. Uncertainty and disturbance estimator-based controllers design under finite control bandwidth constraint. IEEE Trans. Ind. Electron. Feb. 2018;65(2):1439–49.
[39] Aharon I, Shmilovitz D, Kuperman A. Phase margin oriented design and analysis of UDE-based controllers under actuator constraints. IEEE Trans. Ind. Electron. Oct. 2018;65(10):8133–41.
[40] S. Borafker and A. Kuperman, Dual-notch-equipped DC link voltage controller for single-phase grid-connected converters, Accepted for presentation at EPE-ECCE Europe, 04 – 08 Sept. 2023, Aalborg, Denmark.
[41] Yepes A, et al. Effects of discretization methods on the performance of resonant controllers. IEEE Trans. Power Electron. Jul. 2010;25(7):1692–712.
[42] Elkayam M, et al. Loop shaping by single-resonant controllers for prescribed tracking of sinusoidal references. IEEE Trans. Power Electron. Nov. 2019;34(11):11352–60.
[43] https://www.ti.com/tool/TMDSCNCD28335.